Comparative Analysis of CNN, Vision Transformers, and Hybrid Models for Skin Lesion Classification using the HAM10000 Dataset

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ABSTRACT

COMPARATIVE ANALYSIS OF CNN, VISION TRANSFORMERS, AND HYBRID MODELS FOR SKIN LESION CLASSIFICATION USING THE HAM10000 DATASET

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The early and accurate classification of skin lesions is essential for the timely diagnosis and treatment of skin cancers. This thesis presents a comparative study of three deep learning architectures—EfficientNet-B3, Vision Transformer (ViT-Small), and a hybrid model combining both—for multiclass classification of dermoscopic skin lesion images from the HAM10000 dataset. The dataset was preprocessed using a custom pipeline that included hair artifact removal (DullRazor method) and CLAHE-based contrast enhancement. To address class imbalance, targeted data augmentation was applied to underrepresented classes, and class-balanced training was performed using weighted loss functions. All three models were trained and evaluated using a consistent pipeline, with and without test-time augmentation (TTA). The EfficientNet-B3 model achieved the highest accuracy (92.99%), while the ViT-Small model delivered the best macro F1-score (92.62%) on rare lesion types. The hybrid model demonstrated significant improvements when TTA was applied but performed less favorably overall. This work demonstrates that CNNs provide stable performance across all classes, ViTs excel in classifying rare lesions, and hybrid models require more advanced feature fusion techniques to compete. These findings can help guide future research and deployment of AI-driven dermatology tools.

Keywords: Skin lesion classification, EfficientNet, Vision Transformer, hybrid deep learning, HAM10000, test-time augmentation, class imbalance, preprocessing.

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CHAPTER 1

INTRODUCTION

1.1 Background

Skin cancer has emerged as one of the most frequently diagnosed forms of cancer worldwide. Among its types, melanoma poses the highest risk due to its rapid progression and potential to metastasize if not detected early. Early diagnosis remains the most effective way to improve survival rates. According to the World Health Organization, millions of skin cancer cases are reported each year, with early-stage detection significantly improving treatment outcomes.

To support early detection, dermoscopy has become a widely used non-invasive imaging technique. It enhances visualization of subsurface skin structures and improves diagnostic accuracy when compared to naked-eye examination [1]. However, its effectiveness is highly dependent on the experience and skill level of the clinician, often leading to inconsistencies in diagnosis. In areas with limited access to dermatological care, the reliance on expert interpretation becomes a critical barrier.

To address this challenge, computer-aided diagnosis (CAD) systems have gained attention, especially those powered by artificial intelligence (AI) and deep learning (DL) [2, 5]. These systems offer the potential to automate image interpretation, reduce human error, and improve diagnostic reliability. Among deep learning techniques, Convolutional Neural Networks (CNNs) have shown strong performance in analyzing medical images, including dermoscopic data [4, 6], due to their ability to capture local visual patterns.

Despite their success, CNNs have limitations. Their architecture is designed to focus on local features through spatially restricted filters, which may not fully capture global relationships across an image. To overcome this, Vision Transformers (ViTs), originally developed for natural language processing, have been adapted for image classification [27, 9, 10]. ViTs process images as sequences of patches and use self-attention to model long-range dependencies.

More recently, hybrid models combining CNNs and ViTs have been introduced, aiming to harness the strengths of both architectures [12, 13]. These models attempt to retain CNNs' local pattern recognition capabilities while also leveraging transformers' ability to understand global context.

1.2 Motivation

The rapid growth of deep learning in medical imaging has led to significant advancements in automated skin lesion classification. Yet, despite these developments, challenges persist that limit the translation of these models into clinical practice [6, 9].

One of the key issues is the inconsistency in performance across different architectures. Many studies demonstrate impressive results using Convolutional Neural Networks (CNNs) or Vision Transformers (ViTs), but few provide a structured comparison between the two under standardized conditions [10, 34]. Without a uniform experimental setup, it becomes difficult to determine which model architecture is more reliable or generalizable for real-world deployment [12, 13].

Moreover, the effectiveness of any classification model is deeply influenced by the quality of input images. Dermoscopic images often contain visual artifacts, such as hair, shadows, and uneven lighting, which can obscure critical lesion details [23, 19]. These issues are particularly detrimental in models that rely heavily on fine-grained texture information [3, 5]. Despite the availability of preprocessing methods like DullRazor for hair removal and Contrast Limited Adaptive Histogram Equalization (CLAHE) for enhancing local contrast, such techniques are often underutilized or omitted entirely in recent deep learning workflows—especially in transformer-based studies [28].

Another critical factor is class imbalance. In datasets like HAM10000, common lesion types are overrepresented while rarer, clinically significant classes have very few samples [1]. This imbalance can cause models to become biased toward frequent classes, reducing sensitivity to rare but dangerous conditions. Addressing this requires careful balancing strategies, such as targeted data augmentation [29], class-balanced or focal loss functions [13], and label smooth-

ing [12].

Given these gaps, there is a strong need for a systematic and unified study that incorporates:

- standardized preprocessing of dermoscopic images [19, 24],
- balanced training data across all classes [29, 30],
- and a controlled comparison of CNN, ViT, and hybrid architectures [12, 10].

This research is driven by the goal of understanding not just which model performs best, but why, under what conditions, and with what trade-offs. By aligning model training, preprocessing, and evaluation within a consistent pipeline, this study aims to provide actionable insights for developing more reliable skin lesion classifiers.

1.3 Problem Statement

Deep learning has significantly improved the field of skin lesion classification, yet several unresolved challenges continue to limit its effectiveness in clinical environments.

Many existing approaches rely on either convolutional neural networks (CNNs) or transformer-based models to classify dermoscopic images. While CNNs are effective in capturing local visual patterns such as color and texture [4, 6], they often fall short in modeling global context.

Conversely, Vision Transformers (ViTs) excel at understanding long-range dependencies but may overlook important local features unless properly guided or initialized [9, 10].

Despite these complementary strengths, few studies attempt a direct, controlled comparison of CNNs, ViTs, and their hybrid combinations using identical training conditions [12, 13]. Most prior work varies in datasets, preprocessing, model depth, or loss functions, making it difficult to fairly assess which architecture is most suitable for clinical-grade lesion classification.

Additionally, dermoscopic images often contain artifacts like body hair, low contrast, and color distortions. Without proper preprocessing, these visual obstructions can impair model performance, especially for underrepresented or subtle lesion types. Although techniques like DullRazor for hair removal [19, 23] and CLAHE for contrast enhancement [24, 28] exist, they

are frequently omitted in modern transformer-based pipelines, leading to suboptimal feature extraction.

Class imbalance poses another major obstacle. In the HAM10000 dataset, a few lesion types dominate the sample distribution, while other clinically important categories remain severely underrepresented [1]. This imbalance can cause models to overfit to frequent classes, neglecting rare but dangerous lesions. Addressing this issue requires strategies such as targeted data augmentation [29], class-balanced or focal loss functions [13], and label smoothing [12].

Given these limitations, this study addresses the following research problem:

How do different deep learning architectures—EfficientNet-B3 (CNN), ViT-Small (Transformer), and a hybrid CNN-ViT model—compare in classifying dermoscopic skin lesions when trained under a unified preprocessing pipeline and evaluated with proper class balancing and standardized metrics?

This research seeks to bridge the gap between academic experimentation and real-world applicability by building a fair and interpretable framework for model comparison.

1.4 Research Objectives

The primary goal of this research is to explore how different deep learning architectures perform in the task of multiclass skin lesion classification when supported by consistent preprocessing, balanced training, and standardized evaluation protocols. To address the identified gaps, this study sets out the following specific objectives:

1. **To implement a robust preprocessing pipeline** tailored for dermoscopic images, incorporating:
    - Hair artifact removal using the DullRazor algorithm [19], and
    - Contrast enhancement through CLAHE [25] to improve lesion visibility.
2. **To implement and train three distinct deep learning models** using the same dataset and preprocessing strategy:

- EfficientNet-B3 as a CNN-based architecture [4],
- ViT-Small as a transformer-based model [10], and
- A custom hybrid model that integrates convolutional and transformer components [12].

3. **To address dataset imbalance** using a combination of techniques, including:
   - Targeted data augmentation for underrepresented classes,
   - Class-balanced loss functions such as Focal Loss [13], and
   - Label smoothing to prevent overconfidence in dominant class predictions.
4. **To evaluate model performance** on multiple fronts, including:
   - Overall accuracy and macro-averaged F1 score,
   - Per-class metrics to assess sensitivity across lesion types, and
   - Confusion matrices to visualize misclassifications.
5. **To investigate the effect of Test-Time Augmentation (TTA)** by comparing baseline predictions with TTA-enhanced outputs in all three models [34].
6. **To interpret results comparatively** and analyze which model architecture offers the most balanced performance and is better suited for real-world deployment, particularly in low-resource or clinical screening settings.

1.5 Scope of the Study

This research focuses on the classification of dermoscopic skin lesion images using deep learning techniques. The study is based exclusively on the publicly available HAM10000 dataset [1], which contains 10,015 dermoscopic images across seven diagnostic categories. The scope is intentionally limited to image-level classification tasks and does not include lesion segmentation, patient metadata, or temporal analysis.

Model Selection: Only three types of architectures are considered:

- A convolutional model (EfficientNet-B3) [4],
- A transformer-based model (ViT-Small) [9],
- A hybrid model combining CNN and ViT components [12].

These were chosen for their strong baseline performance and complementary architectural properties. No ensemble learning or large-scale pretraining beyond ImageNet is explored.

Preprocessing Pipeline: All images undergo a standardized preprocessing routine that includes:

- Hair removal using the DullRazor algorithm [19],
- Contrast enhancement via CLAHE [25],
- Image resizing and normalization compatible with deep learning models.

This pipeline is consistently applied across all experiments to ensure fair model comparison.

Data Handling: To address class imbalance, the study uses:

- Stratified data splitting for training, validation, and testing,
- Targeted data augmentation for minority classes [29],
- Focal loss and label smoothing during training [13].

No external datasets or metadata (e.g., patient age, sex, or anatomical site) are included in the training process.

Evaluation: Performance is evaluated using standard metrics such as accuracy, precision, recall, and F1 score. Both base and test-time augmented predictions are considered to measure robustness [28]. Analysis is conducted at both the macro and per-class levels to reflect model fairness and sensitivity to rare lesion types.

Deployment Considerations: The goal is not to develop a deployable clinical system but to perform a comparative analysis of model behaviors under a controlled, well-defined pipeline.

The findings are intended to guide future development of CAD tools and improve model transparency for medical applications.

1.6 Thesis Organization

This thesis is structured into six chapters, each serving a specific purpose in documenting the research journey from motivation to conclusion.

- **Chapter 1 – Introduction:** Introduces the research problem, background context, and motivations. It outlines the key challenges in automated skin lesion classification and defines the objectives, scope, and research question addressed in the study.
- **Chapter 2 – Literature Review:** Reviews prior work in deep learning for skin lesion classification. It summarizes key developments in convolutional neural networks, vision transformers, and hybrid models, along with common preprocessing and class balancing techniques used in medical imaging.
- **Chapter 3 – Materials and Methods:** Describes the dataset, image preprocessing pipeline, model architectures, training configurations, and evaluation metrics used in the study. This chapter also details the experimental design ensuring objectivity across all models.
- **Chapter 4 – Results:** Presents the performance outcomes of all three models under identical testing conditions. Results are analyzed both quantitatively and visually using metrics such as accuracy, F1 score, and confusion matrices.
- **Chapter 5 – Discussion and Summary:** Interprets the experimental findings considering the research objectives. It discusses comparative strengths and weaknesses of each model and considers on practical implications and limitations of the study.
- **Chapter 6 – Conclusion and Future Work:** Summarizes the contributions of the research and suggests possible directions for future exploration, including the use of metadata, larger datasets, and deployment strategies for clinical implementation.

CHAPTER 2

LITERATURE REVIEW

2.1 Deep Learning in Skin Lesion Analysis

The advancement of skin lesion classification has been deeply rooted in classical image processing techniques. Early research emphasized handcrafted features—such as texture, color, and shape—for differentiating between lesion types. Notably, Dr. Umbaugh and collaborators explored texture-based analysis and relative color features to distinguish melanoma from other skin conditions [3, 5]. These classical methods served as the foundation for automated lesion analysis, and even addressed preprocessing tasks like hair artifact removal to enhance image clarity [23]. Similarly, Leander et al. [35] developed algorithms that automatically differentiated melanoma and Clark nevus lesions using structural and color-based cues.

Over the past decade, deep learning has revolutionized medical image analysis, offering powerful capabilities in segmentation, classification, and anomaly detection. In dermatology, the rise in skin cancer cases worldwide has necessitated more accessible and accurate diagnostic tools. Automated classification systems powered by deep learning now serve as vital components in clinical decision support systems, particularly for early detection of melanoma and other skin lesions.

One of the foundational breakthroughs in this field was the release of the HAM10000 dataset by Tschandl et al. [1], a large and diverse collection of dermatoscopic images encompassing seven classes of pigmented skin lesions. This dataset has since become a benchmark for evaluating the performance of various classification models.

Convolutional Neural Networks (CNNs) have traditionally dominated image classification tasks due to their ability to learn hierarchical representations from raw image data. Early work on applying CNNs to skin lesion classification demonstrated performance comparable to dermatologists. This success catalyzed widespread research into deeper and more efficient CNN

architectures such as ResNet, DenseNet, and InceptionNet.

Among the recent innovations, EfficientNet has stood out for its ability to balance accuracy and computational efficiency. EfficientNet-B3, in particular, has been shown to outperform baseline CNNs on HAM10000 when combined with effective data augmentation strategies. Mahmud et al. [34] confirmed this in their study, demonstrating superior classification results through image resampling techniques. Ali et al. [4] also emphasized the strength of EfficientNet for multiclass classification in dermatological settings, citing its compound scaling approach that adjusts network depth, width, and resolution in a principled way.

Saba et al. [6] conducted a comparative analysis of different CNN architectures, revealing that both architectural complexity and preprocessing significantly affect classification performance. Lightweight models such as MobileNet have also gained traction, particularly for applications requiring fast inference and low-resource deployment [7].

These developments have laid the groundwork for increasingly sophisticated CAD systems in dermatology, where CNNs continue to be refined and combined with other techniques to improve both performance and generalizability.

2.2 Vision Transformers and Hybrid Architectures

Despite the dominance of CNNs in medical imaging, their inherent limitation lies in the use of local receptive fields, which restrict the model's ability to capture global contextual information. To address this, the Vision Transformer (ViT) architecture was introduced by Dosovitskiy et al., bringing the power of self-attention mechanisms—originally developed for natural language processing—into computer vision tasks. These models divide an image into patches and apply self-attention to model long-range dependencies, allowing for more holistic feature extraction across the entire image.

The application of ViTs in dermatology has grown rapidly. Rezvantalab and Safigholi [9] developed a ViT-based model for skin lesion classification that utilized attention fusion to improve interpretability and performance. Himel et al. [8] combined ViT with the Segment Anything Model (SAM) to achieve high accuracy in both lesion segmentation and classification

tasks. Their results demonstrated ViT's capability to generalize well across dermoscopic image datasets.

A comprehensive survey by Galib et al. [10] highlighted the emerging prominence of ViTs in dermatology and their comparative advantages over CNNs, particularly in handling complex and varied lesion structures. Further supporting evidence was provided by Nahnani et al. [11], who evaluated ViT-B/16 on the HAM10000 dataset and found it to be effective for multiclass classification with minimal architectural tuning.

To leverage the strengths of both architectures, researchers have proposed hybrid models that combine CNN backbones with transformer heads. These architectures aim to benefit from CNNs' strong local feature extraction and transformers' ability to model global relationships. For instance, Patel et al. [13] proposed a hybrid CNN-ViT model optimized using focal loss, yielding significant improvements in classification metrics. Similarly, Lee et al. [12] introduced the SkinEHDLF model, which fused EfficientNet and ViT to deliver robust performance on skin lesion tasks.

The hybrid approach has also been formalized in architectures like ConvNeXtV2 [17], which integrates CNN and transformer modules into a cohesive framework optimized for medical imaging. Ensemble and fusion-based strategies, as shown in studies by Elahi et al. [14] and Huang et al. [16], further reinforce the effectiveness of combining these architectures to improve generalization, particularly in datasets with limited annotated samples.

Given these advancements, our work investigates the comparative performance of EfficientNet-B3, ViT-Small, and a hybrid CNN-ViT model for skin lesion classification on the HAM10000 dataset, aiming to evaluate their respective strengths in a standardized experimental setup.

2.3 Preprocessing: Hair Removal and Contrast Enhancement

Dermoscopic images frequently suffer from various artifacts such as hair, uneven lighting, and low contrast, which can obscure lesion boundaries and reduce classification accuracy. To mitigate these effects, a robust preprocessing pipeline is essential before feeding the data into a deep learning model [23, 25, 26].

2.3.1 Hair Removal

Hair artifacts are among the most common obstructions in dermoscopic images. These thin, dark strands often cross over lesions and interfere with pixel-level analysis, reducing the accuracy of automated classification systems. Early research at SIUE proposed a dedicated hair detection algorithm specifically for melanoma dermoscopy [23], demonstrating the importance of tailored preprocessing techniques in improving downstream analysis.

A widely adopted solution in the field is the DullRazor algorithm, introduced by Lee et al. [19]. This method involves morphological black-hat filtering to detect dark linear structures (hair), followed by bilinear interpolation or inpainting to fill in the removed pixels. To improve robustness across varying hair colors and skin tones, researchers have proposed enhancements such as E-Shaver [20], which incorporates adaptive edge detection, and SharpRazor [22], which uses region-growing and texture-based segmentation for precise removal.

In this study, we employed the DullRazor algorithm for hair artifact removal, which uses morphological filtering and inpainting techniques. Its simplicity and effectiveness made it a suitable choice for balancing computational efficiency with removal quality [19].

2.3.2 Contrast Enhancement

Low contrast between lesions and surrounding skin can hinder model performance by reducing feature clarity. CLAHE (Contrast Limited Adaptive Histogram Equalization) is a powerful method for enhancing local contrast in dermoscopic images. Unlike global histogram equalization, CLAHE operates on small regions or tiles, adjusting the contrast within each tile while limiting amplification of noise.

Moss et al. [25] and Acebuque and Ruiz [24] demonstrated that applying CLAHE specifically to the L (lightness) channel of the LAB color space effectively enhances lesion visibility without distorting the color balance. Gutierrez-Arriola et al. [26] found that using CLAHE as a preprocessing step improved CNN-based classification by enhancing the borders and textures of lesions.

Furthermore, Cheng et al. [28] showed that CLAHE benefits both CNNs and Vision Transformers, leading to improved model generalization and robustness. In our work, CLAHE was applied after hair removal to enhance lesion contrast while minimizing the impact of residual artifacts.

This dual-stage preprocessing—hair removal followed by CLAHE—was consistently applied across all datasets to ensure uniform enhancement and better model performance.

2.4 Handling Class Imbalance

One of the most prominent challenges in training deep learning models for skin lesion classification is the class imbalance inherent in dermoscopic datasets. The HAM10000 dataset, for example, contains a dominant number of images labeled as melanocytic nevi (nv), while other categories such as dermatofibroma (df) or vascular lesions (vasc) are underrepresented [1]. This skewed distribution can bias a model toward majority classes, thereby reducing the model's ability to accurately detect rare but clinically significant cases.

2.4.1 Impact of Class Imbalance

Class imbalance can lead to several issues in training:

- The model may develop a bias toward majority classes, achieving high overall accuracy but low recall for minority classes [26].
- The loss function may not penalize misclassification of minority classes adequately, further compounding performance disparities.
- Gradient updates become dominated by majority class samples, limiting the learning capacity of the network for rare class features [29].

Given these challenges, addressing class imbalance is essential for building clinically applicable and fair AI systems in dermatology.

2.4.2 Augmentation for Minority Classes

Rather than resampling or duplicating images, which can lead to overfitting, this study employed targeted data augmentation specifically for underrepresented classes. Techniques such as random rotation, color jitter, horizontal flipping, and Gaussian noise were applied selectively to expand the effective dataset size for rare lesion types. This strategy aligns with the findings of Zhang et al. [29], who showed that augmentation combined with attention mechanisms improved minority class learning in dermoscopic datasets.

Such augmentation not only increases the volume of minority class samples but also improves the model's robustness to common perturbations observed in dermoscopic imaging. By applying different augmentations to each minority class image during every epoch, the model encounters more variation and learns more generalized patterns.

2.4.3 Class-Balanced Loss Function

To further address imbalance, a class-balanced variant of the cross-entropy loss function was employed. This loss function adjusts the contribution of each class to the total loss based on the inverse frequency of samples. Specifically, the loss weight for a class c is defined as:

$$w_c = \frac{1-\beta}{1-\beta^{n_c}}$$

where n_c is the number of samples in class c, and $\beta \in [0, 1)$ is a hyperparameter controlling the smoothness of weighting. This formulation is based on the concept of the "effective number of samples," as discussed in [29], and helps to downscale the contribution of frequently occurring classes while amplifying the effect of rare classes.

This method was selected over simple inverse frequency weighting or random oversampling due to its better theoretical justification and lower risk of overfitting. It also integrates seamlessly with modern optimizers such as AdamW and learning rate schedulers like Cosine Annealing, both of which are known to stabilize training in imbalanced scenarios [16].

2.4.4 Alternative Approaches in Literature

Adebiyi et al. [30] highlighted the effectiveness of multimodal fusion to mitigate the adverse effects of imbalance in dermatology-related datasets by integrating metadata and image features. Ensemble-based techniques combining EfficientNet and transformer backbones have also been proposed [16] to stabilize learning across classes, albeit at the cost of computational overhead.

Soft attention techniques [29] and focal loss remain widely used alternatives in recent literature, particularly for medical classification tasks where recall on rare cases is prioritized.

2.4.5 Justification for Approach

The combined use of targeted data augmentation and class-balanced loss allowed for effective training of models without relying on external metadata or complex post-processing pipelines. This approach ensured fair performance across all lesion categories and improved generalization on underrepresented classes, which is critical in real-world diagnostic scenarios. Moreover, by avoiding ensemble methods, this strategy maintained a favorable balance between model accuracy and computational efficiency.

2.5 Recent Innovations and Ensemble Approaches

The field of skin lesion classification has seen significant innovation in recent years, particularly with the introduction of advanced model architectures, fusion strategies, and ensemble-based systems. These advancements aim to address challenges such as overfitting, limited dataset size, poor generalization, and performance inconsistency across lesion categories.

2.5.1 Hybrid Architectures

To leverage both local spatial features and global context, hybrid models that combine CNNs and Vision Transformers have gained momentum. For instance, the SkinEHDLF framework [12] merges EfficientNet and ViT modules into a unified system, effectively improving both sensitivity and specificity for challenging dermoscopic classes. These models are particularly useful for capturing subtle patterns in high-resolution skin images where both pixel-level and contex-

tual features matter.

Similarly, Patel et al. [13] introduced a hybrid CNN-ViT pipeline optimized with focal loss, achieving enhanced accuracy while addressing class imbalance and small lesion detection. Hybrid models like these tend to outperform standalone CNNs or ViTs, especially when combined with preprocessing and augmentation strategies.

2.5.2 Transformer-CNN Blends and Modular Attention

Newer variants like ConvNeXt [17], which blend architectural concepts from both CNNs and Transformers, have shown competitive performance with reduced parameter counts and training costs. ConvNeXtV2 adapts hierarchical convolutional structures with transformer-style attention heads, making it suitable for medical datasets with limited size and high class imbalance.

PanDerm-Swin [18] is another notable contribution that uses pre-trained Swin Transformers and fine-tunes them for skin lesion classification. The model leverages window-based attention and transfer learning to achieve strong generalization without requiring very large labeled datasets.

2.5.3 Ensemble Learning for Robustness

While single-model systems have demonstrated strong performance, ensemble learning continues to be a powerful technique for improving robustness and accuracy. Huang et al. [16] proposed an ensemble of EfficientNet variants integrated with attention modules. Their approach boosted macro-average metrics and particularly improved performance on underrepresented lesion types.

However, ensemble models come with trade-offs, including:

- Increased training and inference time.
- Higher computational and memory requirements.
- Difficulties in real-time deployment and edge computing integration.

As a result, while ensembles can achieve state-of-the-art results, their practicality in clinical settings remains limited unless efficiency-focused adaptations are made.

2.5.4 Efficient Implementations

In practical implementations, researchers like Jaycee404 [15] have made publicly available hybrid EfficientNet-ViT models optimized for HAM10000 classification. These implementations allow fast deployment and fine-tuning on local machines without significant cloud resources, encouraging reproducibility and further innovation.

Additionally, several lightweight alternatives such as MobileNet [7] and DenseNet-based hybrids [14] have been tested for their ability to balance performance with speed. While not always achieving the topmost accuracy, they offer efficient deployment options on embedded and mobile devices, crucial for point-of-care diagnostics.

2.5.5 Summary

Recent literature clearly demonstrates that no single architecture universally dominates skin lesion classification. The choice of model must be context-dependent, balancing performance, computational cost, and deployment requirements. Hybrid and ensemble approaches are promising directions but must be evaluated within unified experimental frameworks for meaningful comparison, as undertaken in this study.

2.6 Literature Gap and Scope of this Work

Despite the rapid progress in skin lesion classification using deep learning, several critical gaps persist in the current body of literature.

2.6.1 Lack of Unified Comparative Frameworks

Most existing studies evaluate a single class of models, either CNNs [4, 6] or Vision Transformers [9, 11], using different datasets, preprocessing techniques, and evaluation metrics. Such inconsistencies make it difficult to directly compare the effectiveness of different model types

under identical conditions.

Furthermore, only a limited number of works integrate both CNNs and ViTs into a **hybrid framework** within a **standardized experimental pipeline**. Hybrid models like those proposed in [12, 13] offer promising results but often lack direct comparisons to standalone CNN or ViT architectures under the same preprocessing and training conditions. Without such comparative studies, it is unclear which model class is most appropriate for specific clinical deployment scenarios.

2.6.2 Underutilization of Preprocessing in Transformer Pipelines

While classical CNN pipelines frequently incorporate robust preprocessing methods, such as hair removal using DullRazor [19] or E-Shaver [20], and contrast enhancement via CLAHE [24, 25], these techniques are often neglected in Transformer-based approaches. For instance, many ViT-focused studies apply minimal preprocessing, assuming that attention mechanisms can inherently overcome image noise and artifacts. However, artifacts like hair, shadows, and color inconsistencies still degrade feature quality, particularly in smaller datasets.

This gap calls for a detailed investigation into whether preprocessing techniques improve ViT and hybrid model performance in dermoscopic classification tasks.

2.6.3 Class Imbalance Often Overlooked or Minimally Addressed

Though the HAM10000 dataset is widely used, its skewed class distribution remains a major challenge [1]. While some studies adopt basic augmentation [10] or use standard weighted loss functions, very few leverage **targeted data augmentation for minority classes** in tandem with **advanced loss functions** like class-balanced loss [16] or focal loss [13]. This limits the effectiveness of such models in detecting rare, yet critical, lesion types.

There remains a need for an integrated approach that combines preprocessing, augmentation, and customized loss functions to address this imbalance holistically.

2.6.4 Limited Focus on Generalization and Fairness

Many studies report impressive results on validation sets but fail to generalize well in real-world scenarios due to overfitting, poor augmentation strategies, or lack of cross-validation. Additionally, few works analyze per-class performance metrics in detail, especially for minority categories like vascular lesions or dermatofibromas [34]. Clinical deployment requires not just high average accuracy but consistent and fair performance across all diagnostic classes.

2.6.5 Scope of This Work

To address these gaps, this thesis proposes a unified and rigorous framework for skin lesion classification using:

- Three model architectures: EfficientNet-B3 [4], ViT-Small [9], and a custom Hybrid CNN-ViT [12].
- Preprocessing techniques such as DullRazor [19] for hair removal and CLAHE [25] for contrast enhancement.
- Class imbalance handling through selective augmentation and a class-balanced cross-entropy loss [16].
- Uniform evaluation metrics across all models, including per-class accuracy, macro-averaged F1-score, and confusion matrix analysis.

By standardizing the experimental pipeline and comparing all models under identical conditions, this study aims to offer deeper insights into which model architecture is most suitable for real-world skin lesion classification tasks. Additionally, the focus on underrepresented classes and fairness in evaluation seeks to promote clinically viable and equitable AI applications in dermatology.

2.7 Summary

This chapter reviewed the evolution of deep learning approaches for skin lesion classification, focusing on CNNs, Vision Transformers, and hybrid models. It highlighted key preprocessing methods such as DullRazor and CLAHE, and addressed the importance of handling class imbalance using targeted augmentation and class-balanced loss. Recent innovations in ensemble learning, attention mechanisms, and foundation models were also discussed. These insights informed the design of our pipeline, which combines EfficientNet-B3, ViT-Small, and hybrid architectures under a unified, well-preprocessed, and class-balanced framework.

CHAPTER 3

MATERIALS AND METHODS

3.1 Overview of Methodology

This chapter outlines the experimental framework employed to compare the performance of three deep learning models—EfficientNet-B3, Vision Transformer (ViT-Small), and a hybrid CNN-ViT model—for skin lesion classification using the HAM10000 dataset.

The methodology involves several sequential stages, starting with dataset restructuring into class-specific folders, followed by image preprocessing techniques such as hair artifact removal and contrast enhancement. Stratified splitting and targeted data augmentation were performed to address the significant class imbalance.

Each model was adapted to accept standardized input sizes and trained using identical hyper-parameters, class-aware loss functions, and learning rate schedulers. Test-Time Augmentation (TTA) was also applied to assess the robustness of predictions under stochastic perturbations.

A uniform evaluation protocol using accuracy, macro F1-score, and confusion matrices was maintained across all models to ensure fair comparison and reproducibility. This structured approach facilitates an objective assessment of each architecture's strengths and limitations in handling dermatoscopic image classification.

3.2 Dataset Description

3.2.1 HAM10000 Dataset

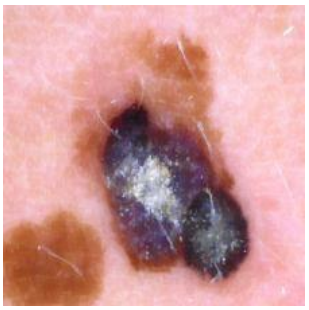
The HAM10000 (Human Against Machine with 10,000 training images) dataset [1] is a widely recognized and extensively used benchmark in the domain of dermatological image analysis. It consists of 10,015 dermatoscopic images collected from various populations and clinical sources, making it one of the most diverse and clinically relevant datasets available for skin

lesion classification research. Each image has a fixed resolution of 600×450 pixels and captures pigmented skin lesions with varying sizes, shapes, colors, textures, and surrounding skin conditions. The images were obtained using different dermatoscopic devices and imaging techniques, thereby introducing variations in illumination, focus, and framing, which are important for training robust machine learning models.

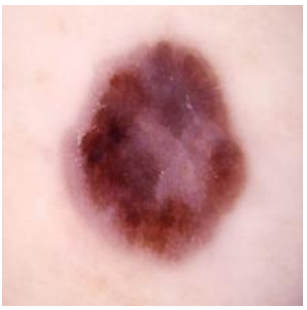
A key strength of the HAM10000 dataset lies in its coverage of seven distinct categories of skin lesions, carefully annotated by expert dermatologists. These categories represent a broad spectrum of clinical presentations, including benign and malignant lesions with different pathological origins. While some lesion types appear frequently in routine screenings, others are rare but clinically significant, making the dataset an ideal benchmark for testing a model's ability to generalize across common and uncommon conditions. Furthermore, the dataset presents a realistic class imbalance, as certain lesion types are overrepresented, while others occur infrequently—mirroring real-world dermatology scenarios. This aspect makes the HAM10000 dataset not only suitable for standard classification tasks but also for evaluating techniques that address data imbalance, such as class weighting, data augmentation, and focal loss.

The seven lesion categories covered in the dataset are visually represented below using sample images from this study. These examples highlight the diversity in lesion morphology, color variation, and surrounding skin texture that a classification model must learn to distinguish.

nv – Melanocytic nevi



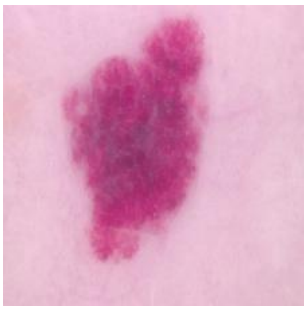
mel – Melanoma



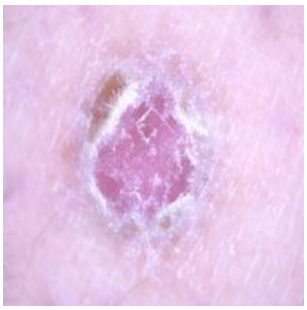
bkl – Benign keratosis-like lesions

bcc – Basal cell carcinoma

akiec – Actinic keratoses / Bowen's disease



vasc – Vascular lesions



df – Dermatofibroma

Figure 3.1: Representative images for each of the seven lesion classes in the HAM10000 dataset.

Descriptions of each class:

- **nv (Melanocytic nevi):** Benign melanocytic tumors; most common lesion type, generally harmless.
- **mel (Melanoma):** Highly malignant and aggressive tumors with potential to metastasize.
- **bkl (Benign keratosis-like lesions):** Includes solar lentigines, seborrheic keratoses, and lichen planus-like keratoses.
- **bcc (Basal cell carcinoma):** A slow-growing but locally invasive form of skin cancer.

- **akiec (Actinic keratoses / Bowen's disease):** Precancerous lesions or early-stage squamous cell carcinoma.
- **vasc (Vascular lesions):** Includes angiomas, angiokeratomas, and hemorrhages.
- **df (Dermatofibroma):** Firm, benign nodules usually resulting from minor trauma or insect bites.

Each image is 600×450 pixels in JPEG format and is accompanied by metadata like patient age, gender, and lesion location. In this study, we used only the image data for classification.

3.2.2 Data Organization and Class Distribution

Originally, the HAM10000 dataset was provided with images in a single directory and annotations in a separate CSV file. For effective data handling and class-wise augmentation, the dataset was reorganized into class-specific folders. Each lesion class had its own folder containing all corresponding images, allowing more efficient preprocessing and batch loading.

The dataset exhibits a highly imbalanced class distribution. This poses significant challenges in training deep learning models, especially in achieving good performance on rare classes. The original class distribution is presented in Table 3.1.

Table 3.1: Original class distribution of the HAM10000 dataset.

Lesion Class	**Number of Images**
Melanocytic nevi (nv)	6705
Melanoma (mel)	1113
Benign keratosis-like lesions (bkl)	1099
Basal cell carcinoma (bcc)	514
Actinic keratoses (akiec)	327
Vascular lesions (vasc)	142
Dermatofibroma (df)	115

The dataset is heavily dominated by the *nv* class, which comprises more than 65% of the total images. In contrast, classes such as *df* and *vasc* have fewer than 150 samples, making them significantly underrepresented. This imbalance can cause deep learning models to become biased toward frequent classes, ignoring minority ones during training.

To address this issue, augmentation techniques were later applied to balance the training set, as described in Section 3.3. The balanced data ensured that each class had approximately equal representation, improving the model's ability to learn subtle features across all classes.

3.3 Data Reorganization, Stratified Splitting, and Augmentation

To prepare the HAM10000 dataset for deep learning classification, we followed a multi-step strategy that involved reorganizing the data, addressing class imbalance, and applying consistent splitting and augmentation. Each step was performed in a specific order to ensure the quality and fairness of the experimental results.

Step 1: Reorganizing Data into Class-Specific Folders

Initially, all images in the HAM10000 dataset were stored in a single folder, with class labels provided through a metadata CSV file. To streamline class-wise preprocessing, augmentation, and loading via PyTorch's `ImageFolder` API, we reorganized the dataset into separate subdirectories for each of the seven skin lesion classes: *nv*, *mel*, *bkl*, *bcc*, *akiec*, *vasc*, and *df*. Each image was moved into the corresponding class folder based on its diagnosis label from the CSV file.

This reorganization allowed us to:

- Perform class-specific augmentation operations
- Easily visualize class distributions
- Load and balance data using standard tools

Step 2: Identifying and Addressing Class Imbalance

Upon reorganizing, it became apparent that the dataset was highly imbalanced. For instance, the class *nv* had 6705 images, while *df* and *vasc* had only 115 and 142 images respectively. This imbalance can significantly degrade model performance, especially for minority classes, since neural networks tend to be biased toward majority class predictions.

To mitigate this, we employed a class-balancing strategy by oversampling the minority classes through augmentation. Using the Albumentations library, we generated synthetic images for each underrepresented class until all classes had approximately 1000 images. This ensured a uniform representation of all lesion categories in the training data.

Step 3: Splitting the Dataset with Stratification

After balancing the data at the folder level, we performed a stratified split of the entire dataset into three non-overlapping subsets:

- **Training set (70%)** – Used for model learning
- **Validation set (15%)** – Used for hyperparameter tuning and early stopping
- **Test set (15%)** – Used for final performance evaluation

Stratification ensured that the relative class proportions were maintained in each subset, which is especially important when working with originally imbalanced data. This helped avoid introducing bias during training or evaluation.

Step 4: Augmentation of Training Data Only

To improve generalization and reduce overfitting, we applied data augmentation to the training set only. The validation and test sets were kept untouched to serve as a true reflection of model performance.

The following transformations were applied:

- **Flipping:** Random horizontal and vertical flips to simulate orientation variations
- **Rotation:** Random rotations between -30° and $+30^\circ$ to simulate camera angles

- **Cropping and Padding:** To introduce size and zoom variations
- **Brightness and Contrast Adjustments:** To simulate lighting variations
- **Gaussian Blur and Sharpening:** To enhance or suppress fine details
- **Gaussian Noise Injection:** To simulate sensor noise and improve robustness

The augmentation intensity was higher for classes that originally had fewer samples, ensuring diversity in synthetic images while preserving lesion morphology.

Outcome and Justification

This combination of folder-based organization, class-specific balancing, stratified data splitting, and controlled augmentation provided a robust and fair experimental setup. By ensuring that each class had equal representation in the training set and was fairly evaluated in the validation and test sets, we avoided bias and made the performance comparison between models reliable and reproducible.

3.4 Preprocessing Pipeline

Effective preprocessing plays a crucial role in medical image analysis, particularly for skin lesion classification, where visual quality and consistency are vital for robust model performance. This section outlines the preprocessing steps applied to the HAM10000 dataset to ensure uniformity, enhance lesion features, and minimize the influence of irrelevant artifacts.

3.4.1 Hair Artifact Removal using DullRazor

Dermatoscopic images in HAM10000 often contain unwanted artifacts such as hairs, ruler marks, and skin folds. These artifacts can obscure lesion boundaries and introduce noise into the learning process, leading to suboptimal model performance.

To address this, the classical **DullRazor algorithm** proposed by Lee et al. [19] was employed. This method is well-established in dermatology image preprocessing and involves the following steps:

1. **Morphological Black-Hat Filtering:** Used to detect dark linear structures such as hairs by subtracting the closing of the image from the original.
2. **Binary Mask Generation:** Adaptive thresholding was applied to the filtered image to segment probable hair regions.
3. **Inpainting:** The segmented hair pixels were removed and the gaps filled using interpolation from neighboring pixels to preserve texture continuity.

This process was applied before any augmentation or contrast enhancement to avoid propagating noise or introducing bias in later stages.

3.4.2 Contrast Enhancement using CLAHE

Skin lesions often suffer from poor lighting and low contrast, making it difficult for models to differentiate subtle features such as edges, color gradients, or internal textures.

To enhance the local contrast without over-amplifying noise, we applied **Contrast Limited Adaptive Histogram Equalization (CLAHE)** [25]. Unlike global histogram equalization, CLAHE operates on small tiles and redistributes the intensities locally, which is beneficial in medical imaging where regional differences matter.

The following steps were performed:

- Convert the RGB image to the LAB color space.
- Apply CLAHE to the L (lightness) channel only.
- Merge the processed L-channel with original A and B channels and convert back to RGB.

This method improved lesion visibility, especially in dark, shadowed, or unevenly lit areas.

3.4.3 Image Resizing and Normalization

To ensure consistency in input size across all models, all images were resized with aspect ratio preserved and padded to 224×224 pixels using bilinear interpolation. Although EfficientNet-

B3 is typically trained with 300×300 input dimensions, a smaller size was chosen to maintain computational feasibility and parity across models.

Next, image intensities were normalized using ImageNet statistics (commonly used for pre-trained models), defined as:

$$\mu = [0.485, 0.456, 0.406], \quad \sigma = [0.229, 0.224, 0.225]$$

This scaling helped align the image distribution with the pre-trained convolutional backbones used in the CNN and hybrid models, ensuring stable gradient propagation and better convergence.

3.4.4 Integration in the Data Pipeline

The entire preprocessing pipeline was implemented using the `OpenCV` and `Albumentations` libraries in Python. Images were preprocessed in batches and saved for reproducibility, then loaded using PyTorch's `DataLoader` during training and testing.

Summary and Justification

- Hair removal was crucial to eliminate non-lesion artifacts that could confuse the model.
- CLAHE enhanced lesion boundary visibility, aiding feature extraction.
- Resizing and normalization aligned data format with model expectations, enabling transfer learning with stable training behavior.
- The preprocessing steps were chosen based on best practices from dermatological image analysis literature [1, 19, 25].

This structured preprocessing ensured that lesion-related features were emphasized, artifacts suppressed, and input consistency maintained, ultimately leading to improved and fairer classification outcomes.

3.5 Model Architectures

This study uses three different deep learning models to classify skin lesion images: EfficientNet-B3, Vision Transformer (ViT-Small), and a custom Hybrid CNN-ViT model. Each model learns features from dermoscopic images in a different way. The goal is to understand which approach is best suited for multiclass classification of skin lesions in the HAM10000 dataset [1].

3.5.1 EfficientNet-B3

EfficientNet is a family of convolutional neural networks (CNNs) that are designed to be both accurate and efficient. Traditional CNNs can be improved by increasing their depth (number of layers), width (number of channels), or input resolution (image size). However, scaling only one of these may not always improve performance.

EfficientNet uses a method called **compound scaling**, which increases depth, width, and resolution together using a balanced formula:

$$d = \alpha^{\varphi}, \quad w = \beta^{\varphi}, \quad r = \gamma^{\varphi} \tag{3.1}$$

Where:

- d is the network depth
- w is the network width
- r is the input resolution
- φ is a user-specified scaling coefficient
- α, β, and γ are constants

The version used in this study, EfficientNet-B3, is one of the mid-sized models in the family, offering good accuracy without requiring high computational power.

MBConv Blocks: EfficientNet uses a special building block called **MBConv** (Mobile Inverted Bottleneck Convolution). It includes:

- **Depthwise separable convolution**: reduces computation
- **Squeeze-and-Excitation (SE) module**: gives more importance to important channels
- **Residual connections**: helps gradient flow in deep networks

Final Layer Modification: The last layer in EfficientNet-B3 was changed to output seven class scores (one for each type of skin lesion in HAM10000), using a softmax activation function.

Table 3.2: EfficientNet-B3 Configuration

Input Image Size	$224 \times 224 \times 3$
Backbone Block	MBConv + SE attention
Number of Parameters	$\sim$12 million
Activation Function	Swish
Pretraining	ImageNet weights
Output Layer	Fully connected (7 classes) + Softmax

3.5.2 Vision Transformer (ViT-Small)

Transformers were originally developed for text processing but have recently shown great success in image classification. The Vision Transformer (ViT) is a deep learning model that breaks an image into small square patches and processes them like a sequence of words.

Step 1: Split Image into Patches An input image of size $224 \times 224 \times 3$ is divided into non-overlapping patches of size 16×16. This gives:

$$N = \frac{224 \times 224}{16 \times 16} = 196 \text{ patches}$$

Step 2: Flatten and Embed Each Patch Each $16 \times 16 \times 3$ patch is flattened into a vector and passed through a linear layer to get a fixed-size embedding (e.g., 384 dimensions).

Step 3: Add Positional Embeddings Because transformers have no concept of position, we add a positional encoding E_{pos} to each embedded patch:

$$z_0 = [x^1E; x^2E; \ldots; x^NE] + E_{\text{pos}}$$

Step 4: Transformer Encoder These embeddings are passed through multiple layers of:

- **Multi-head self-attention (MHSA)**: compares each patch with every other
- **Feed-forward neural networks**
- **Normalization and residual connections**

Step 5: Classification Head A special ”class token” is used to collect global information. After transformer layers, it is passed to a fully connected layer that outputs 7 class probabilities.

Table 3.3: ViT-Small Configuration

Input Size	$224 \times 224 \times 3$
Patch Size	16×16
Number of Patches	196
Embedding Dimension	384
Transformer Layers	12
Attention Heads	6
Output Layer	Fully connected (7 classes) + Softmax

ViTs are very powerful but usually need a lot of training data. They are good at capturing global features, which is useful for complex lesions with large structures.

3.5.3 Hybrid CNN-ViT Model

This model combines both convolutional and transformer-based approaches. The idea is to use the CNN part to extract detailed local features like edges and textures, and the transformer part to understand global structures across the whole image.

Step 1: CNN Feature Extraction The input image is passed through a few convolutional

layers (adapted from EfficientNet-B0) to get a smaller feature map:

$$F \in \mathrm{R}^{C \times H' \times W'}$$

Step 2: Convert Feature Map to Tokens The feature map is flattened into a sequence of tokens. Each spatial position becomes one token:

$$Z_0 = \text{Flatten}(F)E + E_{\text{pos}}$$

Step 3: Transformer Encoding These tokens are passed through transformer layers (fewer than ViT-Small) to learn relationships across different parts of the image.

Step 4: Final Classification The output is pooled and passed through a dense layer that outputs the predicted class probabilities.

Table 3.4: Hybrid CNN-ViT Configuration

Input Size	$224 \times 224 \times 3$
CNN Backbone	EfficientNet-B0 (partial)
Feature Map Shape	$64 \times 14 \times 14$
Token Dimension	384
Transformer Layers	6
Output Layer	Fully connected (7 classes) + Softmax

This hybrid model tries to balance the strengths of both CNNs and transformers. It is particularly helpful when local texture and global shape are both important for classification.

3.6 Training Strategy

3.6.1 Loss Function

To effectively handle class imbalance present in the HAM10000 dataset, **Focal Loss** was used as the primary loss function. Unlike traditional cross-entropy, focal loss down-weights the

contribution of well-classified examples and focuses training on hard-to-classify cases. It is defined as:

$$L_{\text{focal}} = -\alpha_t(1 - p_t)^{\gamma} \log(p_t)$$

Where:

- p_t is the predicted probability for the true class,
- α_t is a weighting factor to balance class frequencies,
- γ is the focusing parameter (set to 2 in our experiments).

3.6.2 Optimizer and Learning Rate Scheduler

All models were trained using the **AdamW** optimizer, which adds decoupled weight decay to Adam, improving generalization. The initial learning rate was set to 3×10^{-4}.

A **Cosine Annealing with Warm Restarts** scheduler was employed to allow the learning rate to decrease smoothly and periodically restart, improving convergence and helping the model escape local minima. Each cycle began with a higher learning rate and gradually decreased.

3.6.3 Regularization Techniques

To prevent overfitting and improve generalization, several regularization strategies were implemented:

- **Dropout:** A dropout layer with $p = 0.3$ was added before the final classifier in all models.
- **Label Smoothing:** Label smoothing with $\varepsilon = 0.1$ was applied during training to prevent the model from becoming overconfident and to improve calibration.
- **Early Stopping:** Training was halted if validation F1 score did not improve for 20 consecutive epochs.

3.7 Test-Time Augmentation (TTA)

To increase the robustness and reliability of model predictions, Test-Time Augmentation (TTA) was employed during inference. TTA involves generating several augmented versions of each test image—such as horizontal flips, brightness adjustments, and slight rotations—and passing each version through the trained model.

The final prediction for each test image was obtained by one of the following fusion strategies:

- **Probability-level fusion:** Averaging the softmax outputs of all augmented variants.
- **Majority voting:** Selecting the class with the most frequent prediction across all variants.

TTA has been shown to reduce prediction variance and improve generalization, especially for visually ambiguous or borderline lesion cases [29]. In our experiments, it consistently yielded more stable results compared to single-pass inference.

3.8 Experimental Design for Fair Comparison

To ensure a consistent and unbiased evaluation of all three deep learning models, this study followed a carefully standardized experimental framework. The primary goal was to isolate the effect of model architecture on performance while minimizing external variability. The following strategies were employed:

- **Stratified Splitting:** The HAM10000 dataset was divided into training (70%), validation (15%), and test (15%) sets using stratified sampling to preserve class-wise distribution across all subsets.
- **Uniform Preprocessing:** All models received identically preprocessed input images, including hair removal, contrast enhancement (CLAHE), resizing to a fixed resolution, and normalization to match model-specific input distributions.
- **Class Balancing via Augmentation:** Underrepresented classes were augmented using

Albumentations transformations until approximately 1000 samples per class were available, reducing class imbalance and improving training stability.

- **Fixed Hyperparameters:** Training settings such as batch size, learning rate, number of epochs, optimizer (AdamW), scheduler (CosineAnnealingWarmRestarts), and loss function (Focal Loss with label smoothing) were kept consistent for all models.
- **Unified Framework and Platform:** All models were implemented using PyTorch and trained using Google Colab with GPU acceleration. Local runtime support was also enabled to utilize an NVIDIA RTX 4070 GPU (12 GB VRAM) and 32 GB RAM for more resource-intensive experiments.

This experimental setup ensured that architectural differences were the primary variables influencing the observed results. Such rigor promotes reproducibility and strengthens the reliability of performance comparisons.

3.9 Evaluation Metrics

To measure the performance of each model comprehensively, the following metrics were used:

- **Accuracy:** The overall percentage of correctly classified images:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}$$

- **F1 Score:** Harmonic mean of precision and recall. Macro-averaged F1 was used to give equal importance to all classes regardless of their size:

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

- **Confusion Matrix:** Displays true vs. predicted labels across all classes. Useful for identifying patterns of misclassification.

- **Per-Class Metrics:** F1 score, precision, and recall were also calculated for each class to understand class-specific performance, especially important due to class imbalance.
- **TTA vs. Non-TTA Comparison:** Metrics were compared with and without TTA to analyze its effect on performance.

3.10 Summary

This chapter presented a complete overview of the experimental setup used to train and evaluate deep learning models for skin lesion classification. The methodology included class-balanced stratified splitting, data augmentation for underrepresented classes, regularized training with focal loss and label smoothing, and a unified evaluation protocol using test-time augmentation. All models were trained under controlled, identical conditions to ensure a fair and reproducible comparison. The next chapter presents the results obtained from each model and discusses their relative strengths and limitations.

CHAPTER 4

RESULTS

4.1 Overview of Experimental Setup

This chapter presents the results of our comparative analysis of three deep learning models—EfficientNet-B3 (CNN), ViT-Small (Transformer), and a Hybrid CNN-ViT architecture—for skin lesion classification using the HAM10000 dataset. The performance of each model was assessed on a consistent and balanced test set using multiple evaluation metrics, including overall accuracy, macro-averaged F1 score, and class-wise F1 scores. To enhance robustness, all models were also evaluated with and without Test-Time Augmentation (TTA). The results presented here are organized to provide a comprehensive comparison of learning behavior, classification accuracy, and error distribution across lesion types.

The figures and plots referenced in this chapter were generated during model training and evaluation using the PyTorch framework. All models were trained under identical hyperparameter settings, with early stopping and cosine learning rate scheduling applied to ensure optimal convergence. The goal of this analysis is not only to identify the most effective model but also to highlight the strengths and weaknesses of each architecture in the context of medical image classification.

4.2 Training and Validation Performance

This section presents and analyzes the training and validation accuracy and loss curves for the three models: EfficientNet-B3, Vision Transformer (ViT-Small), and the Hybrid CNN-ViT architecture. These curves provide insight into the learning dynamics, convergence behavior, overfitting, and generalization capabilities of each model.

4.2.1 EfficientNet-B3 Training Curves

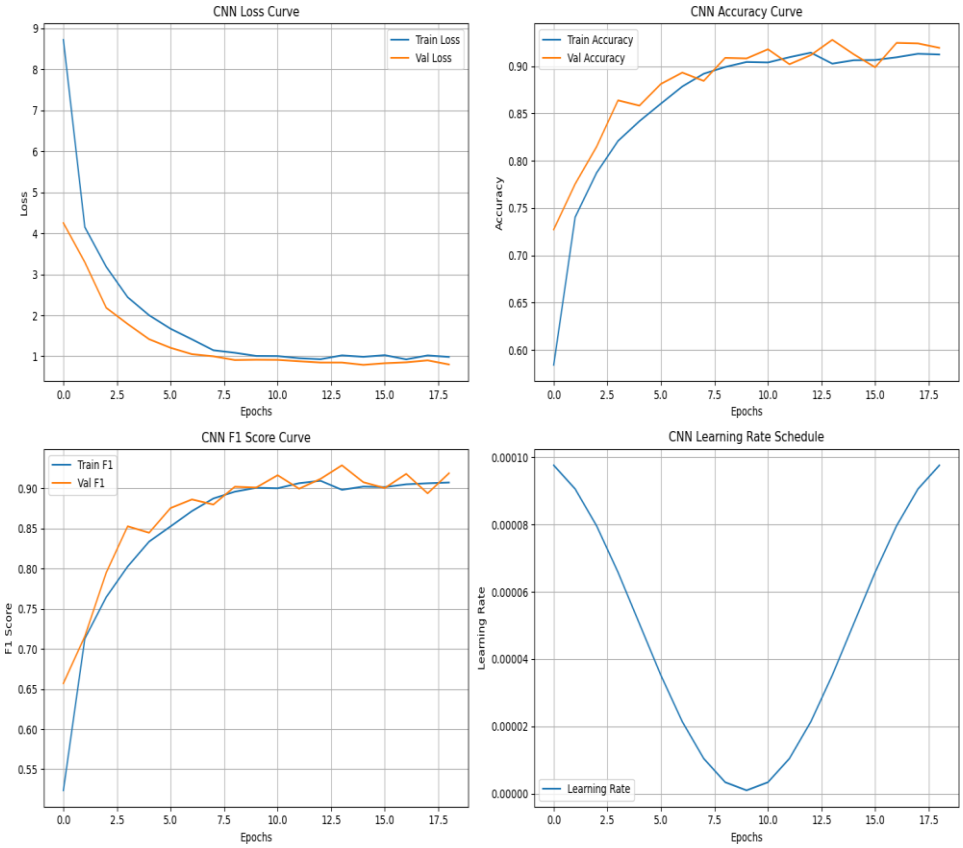


Figure 4.1: Training and Validation Accuracy/Loss Curves for EfficientNet-B3

As shown in Figure 4.1, EfficientNet-B3 demonstrates smooth convergence. The training accuracy steadily increases while the loss decreases, indicating effective learning. The validation accuracy closely follows the training curve, and the validation loss plateaus after about 30 epochs. There is minimal divergence between training and validation curves, suggesting low overfitting and good generalization. The early stopping mechanism halted training at the point of optimal validation accuracy.

4.2.2 ViT-Small Training Curves

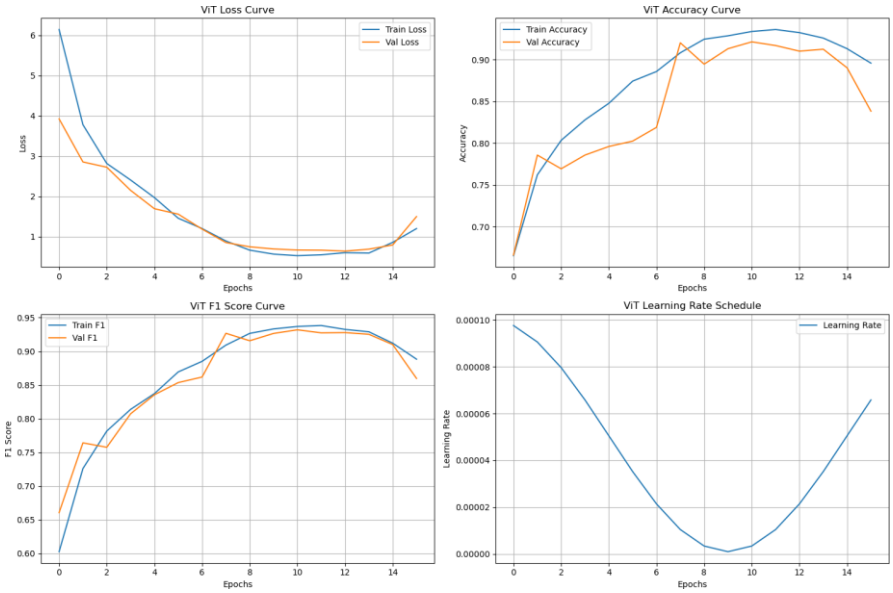


Figure 4.2: Training and Validation Accuracy/Loss Curves for ViT-Small

In Figure 4.2, the ViT-Small model shows slightly noisier training behavior. Initially, the model struggles to learn effectively, with oscillations in both accuracy and loss. However, after epoch 15, the curves begin to stabilize. The validation accuracy improves consistently, although the gap between training and validation performance is wider than in EfficientNet, suggesting moderate overfitting. Despite this, the model shows potential for strong generalization with further tuning.

4.2.3 Hybrid CNN-ViT Training Curves

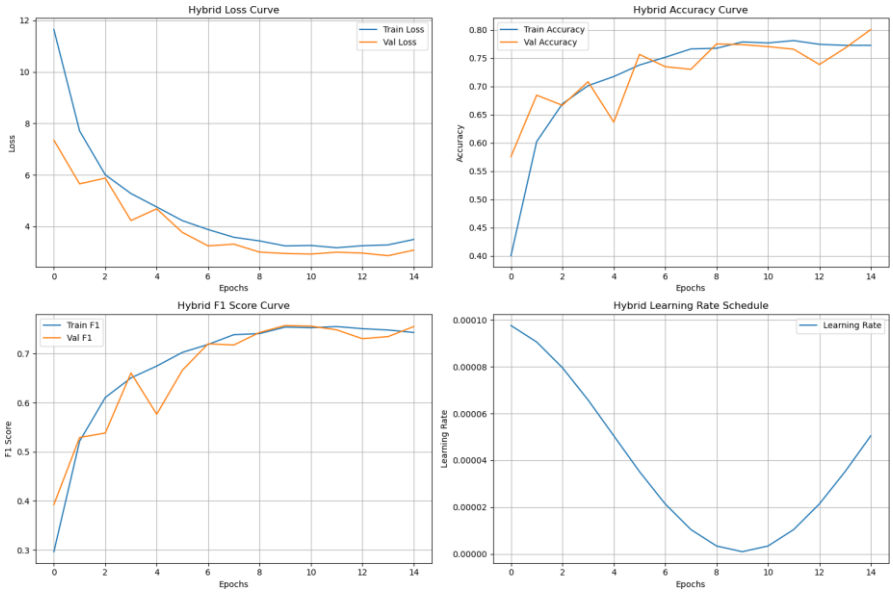


Figure 4.3: Training and Validation Accuracy/Loss Curves for the Hybrid CNN-ViT Model

Figure 4.3 presents the training behavior of the Hybrid model. It achieves rapid gains in training accuracy early on, but the validation loss increases slightly in later epochs. This divergence indicates potential overfitting. The hybrid architecture seems to memorize training patterns quickly but struggles with generalization, likely due to the complexity of integrating CNN and transformer components. Regularization techniques such as dropout and label smoothing were applied, but further tuning may be necessary to improve robustness.

4.3 Confusion Matrix Analysis

The confusion matrix offers a granular view of classification results by comparing the predicted class labels against the true class labels for each of the seven skin lesion categories. Each cell in the matrix represents the number of test samples that belong to the actual class (rows) and were predicted as a specific class (columns). The diagonal elements indicate correct predictions, while off-diagonal elements indicate misclassifications. This section presents the confusion matrices for each model with and without Test-Time Augmentation (TTA).

4.3.1 EfficientNet-B3

EfficientNet-B3 serves as a baseline convolutional model known for its balanced depth, width, and resolution scaling. The confusion matrix for EfficientNet-B3 without TTA reveals the following:

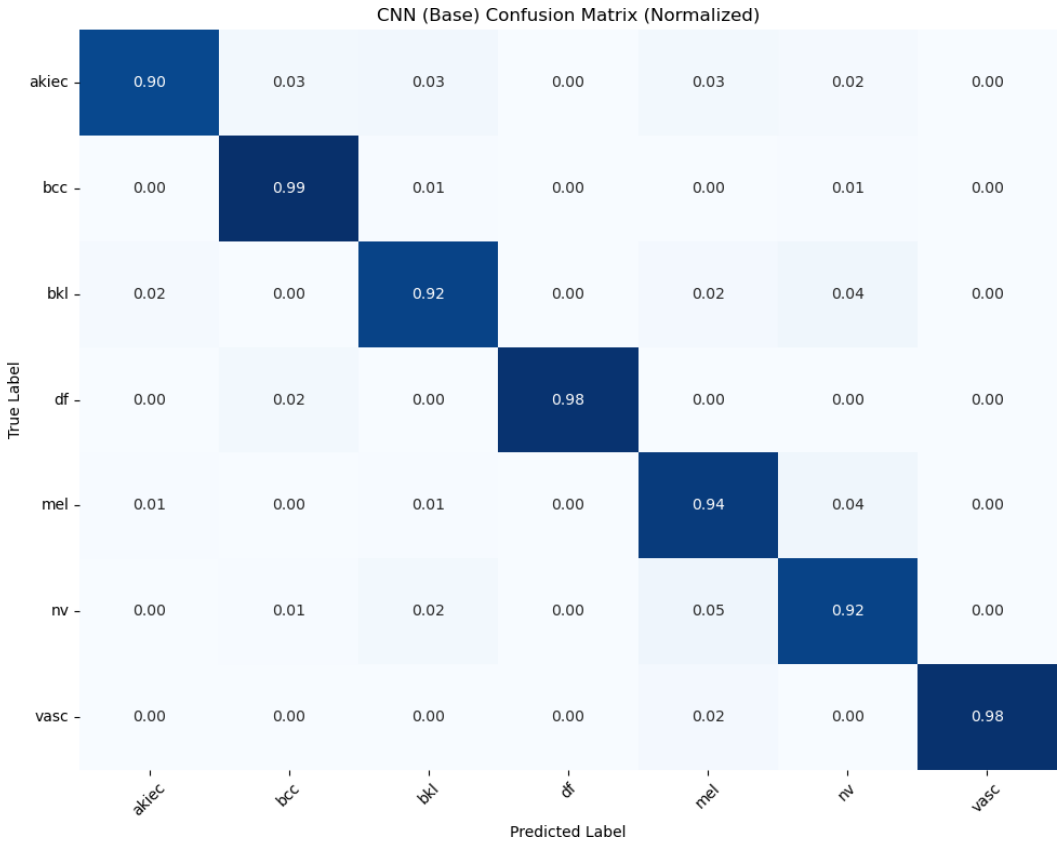


Figure 4.4: Confusion Matrix of EfficientNet-B3 (Without TTA)

Class labels: akiec – Actinic keratoses, bcc – Basal cell carcinoma, bkl – Benign keratosis-like lesions, df – Dermatofibroma, mel – Melanoma, nv – Melanocytic nevi, vasc – Vascular lesions.

- The model classifies nv with high accuracy, as expected due to its large representation in the dataset.
- Misclassifications are notable between bkl and nv, and between mel and bkl, suggesting visual similarity among these classes.
- Classes like df, vasc, and akiec show poorer performance, with frequent confusion

with visually dominant classes.

When TTA is applied:

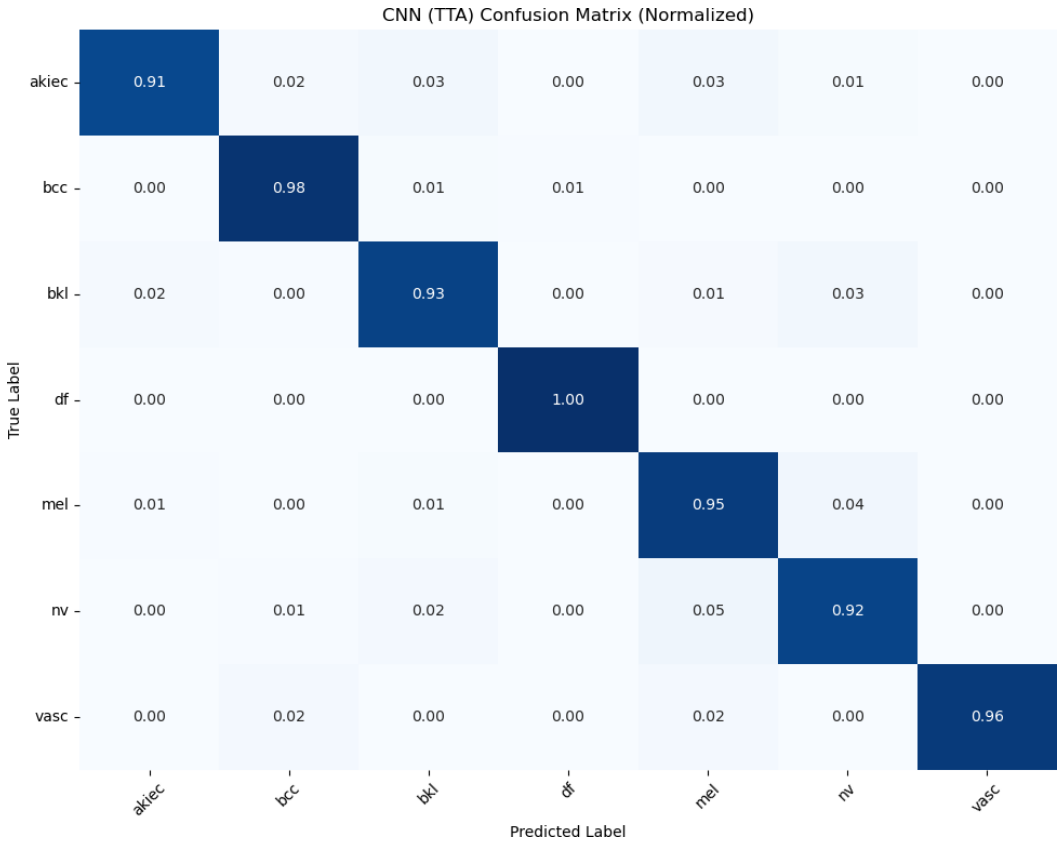


Figure 4.5: Confusion Matrix of EfficientNet-B3 (With TTA)

Class labels: akiec – Actinic keratoses, bcc – Basal cell carcinoma, bkl – Benign keratosis-like lesions, df – Dermatofibroma, mel – Melanoma, nv – Melanocytic nevi, vasc – Vascular lesions.

- Minor improvements are observed across all classes, especially for underrepresented ones like akiec and bcc.
- TTA introduces robustness by averaging predictions from multiple transformed versions of the same input, reducing erratic misclassifications.
- True positive counts for mel and df increase, though false positives still occur.

4.3.2 ViT-Small

The Vision Transformer (ViT-Small) takes a fundamentally different approach by tokenizing image patches and applying self-attention mechanisms. Its base confusion matrix shows:

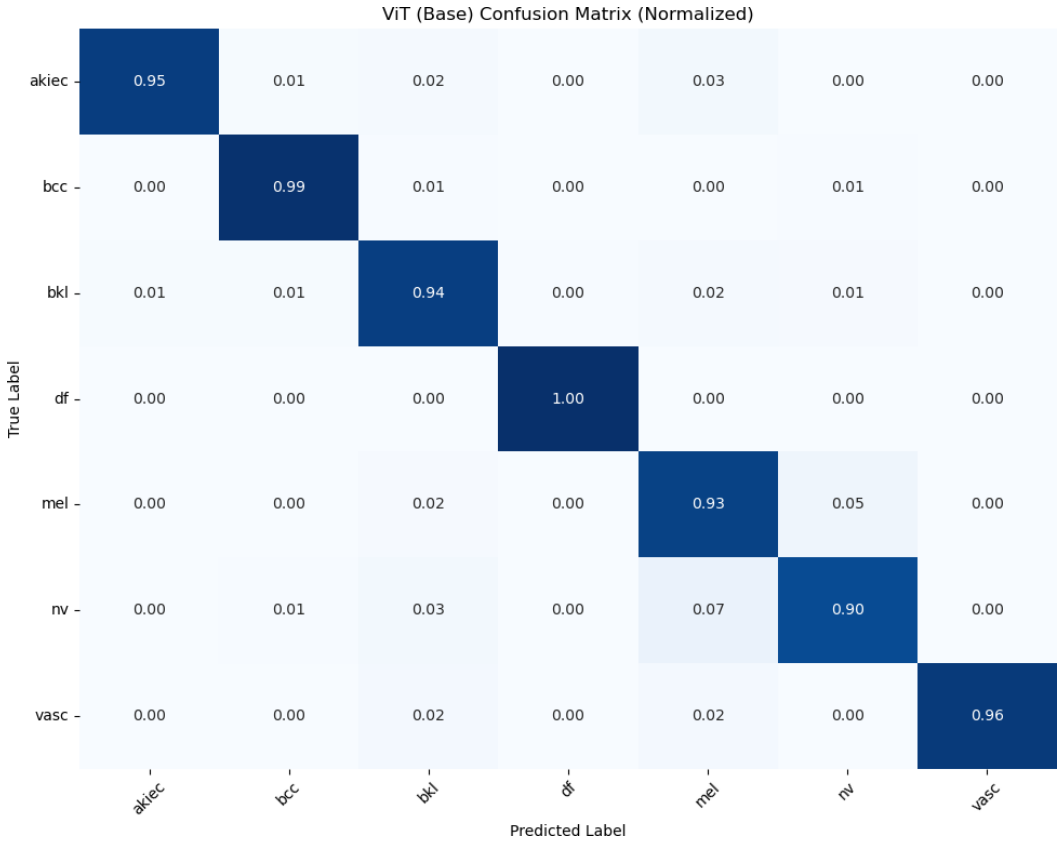


Figure 4.6: Confusion Matrix of ViT-Small (Without TTA)

Class labels: akiec – Actinic keratoses, bcc – Basal cell carcinoma, bkl – Benign keratosis-like lesions, df – Dermatofibroma, mel – Melanoma, nv – Melanocytic nevi, vasc – Vascular lesions.

- ViT performs comparatively better in distinguishing mel from nv, indicating its effectiveness in capturing global contextual cues.
- Similar to EfficientNet, ViT struggles with df and vasc, likely due to their small sample size and low intra-class variance.
- Some misclassifications between bkl and akiec suggest the model may overfit minor details without enough regularization.

TTA significantly stabilizes these results:

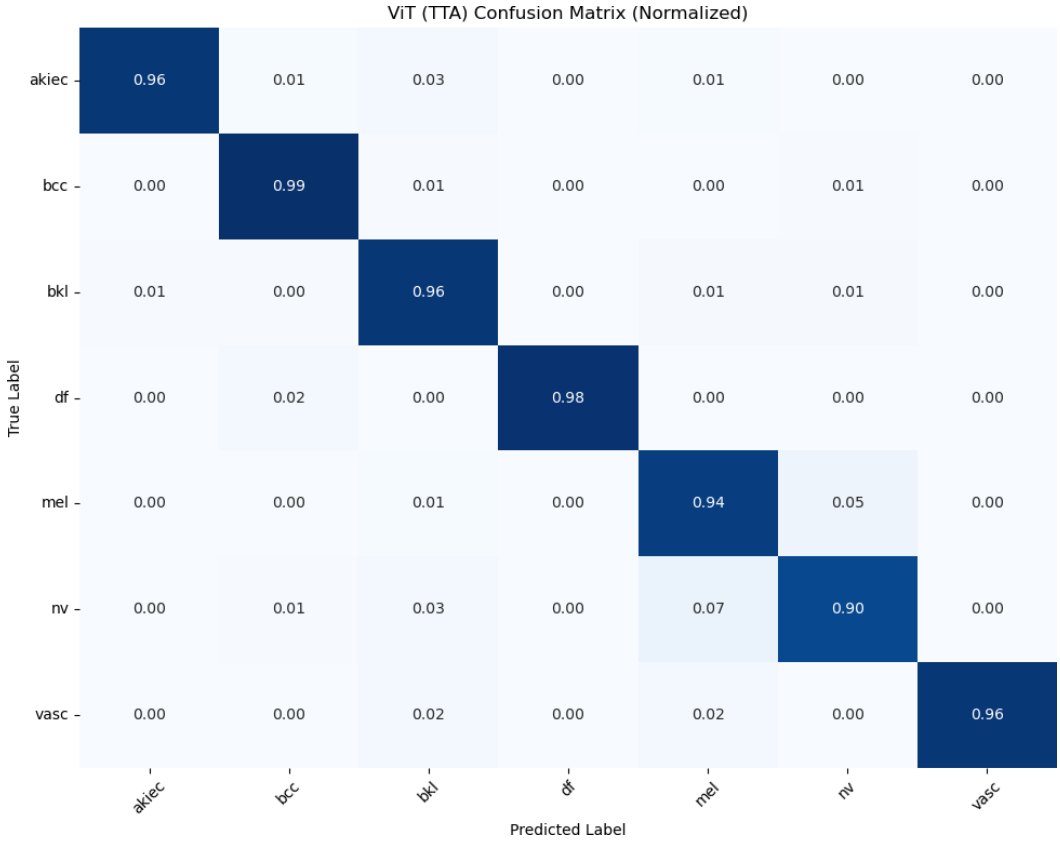


Figure 4.7: Confusion Matrix of ViT-Small (With TTA)

Class labels: akiec – Actinic keratoses, bcc – Basal cell carcinoma, bkl – Benign keratosis-like lesions, df – Dermatofibroma, mel – Melanoma, nv – Melanocytic nevi, vasc – Vascular lesions.

- Greater class separation is achieved, especially in bcc, akiec, and vasc, where ViT earlier showed higher misclassification.
- The matrix becomes closer to a diagonal structure, indicating more correct classifications after averaging predictions over multiple augmented test inputs.
- This improvement reinforces ViT's sensitivity to orientation and spatial representation, which benefits from TTA.

4.3.3 Hybrid CNN-ViT

The hybrid model combines the spatial locality of CNNs with the global feature capturing ability of ViTs. Its base confusion matrix reveals:

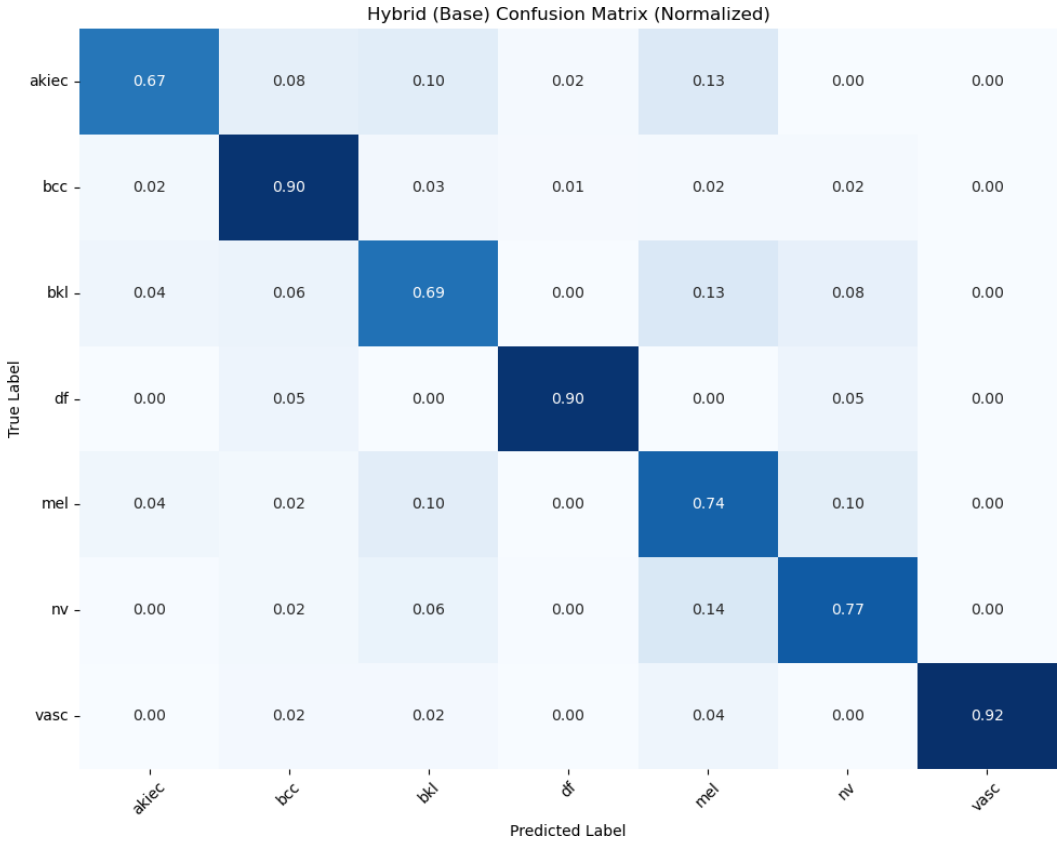


Figure 4.8: Confusion Matrix of Hybrid CNN-ViT (Without TTA)

Class labels: akiec – Actinic keratoses, bcc – Basal cell carcinoma, bkl – Benign keratosis-like lesions, df – Dermatofibroma, mel – Melanoma, nv – Melanocytic nevi, vasc – Vascular lesions.

- Strong performance in majority classes like nv and mel, but weaker in minority ones such as df and vasc.
- More aggressive misclassifications observed compared to EfficientNet and ViT, possibly due to architectural complexity leading to overfitting or poor generalization.
- Confusion between bkl and mel is still present, reflecting limitations in resolving fine-grained lesion types.

With TTA, improvements are visible but modest:

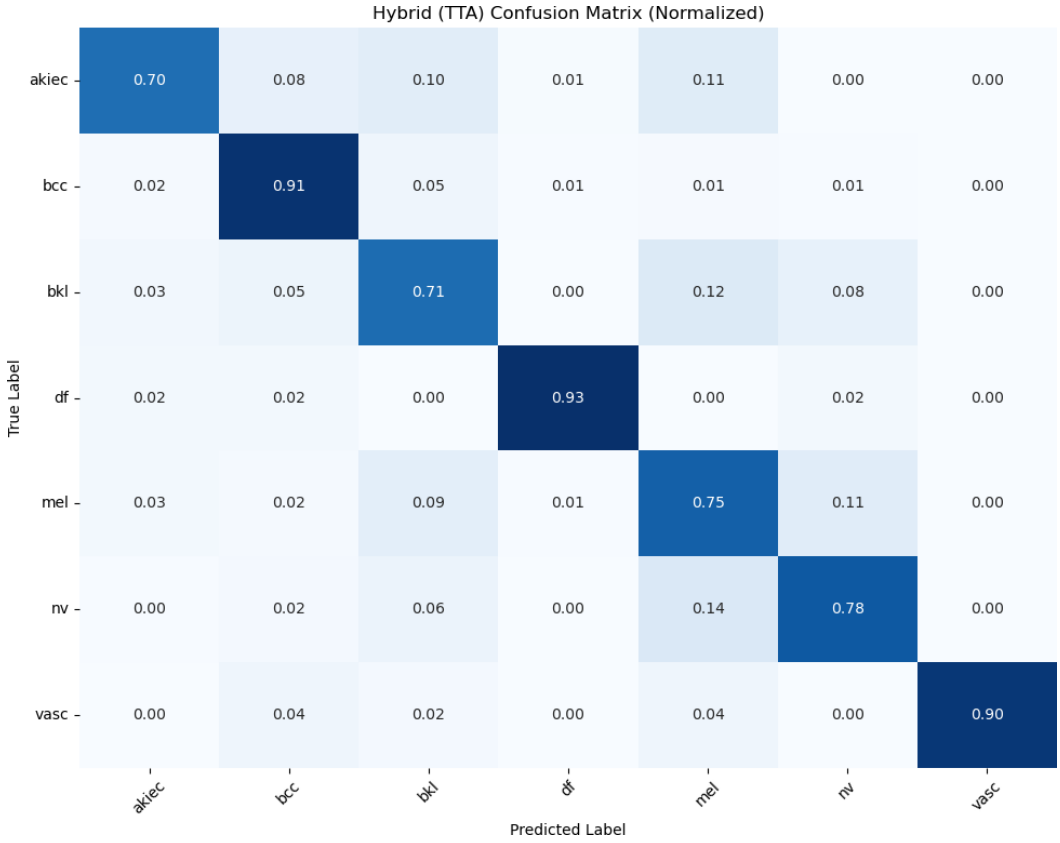


Figure 4.9: Confusion Matrix of Hybrid CNN-ViT (With TTA)

Class labels: akiec – Actinic keratoses, bcc – Basal cell carcinoma, bkl – Benign keratosis-like lesions, df – Dermatofibroma, mel – Melanoma, nv – Melanocytic nevi, vasc – Vascular lesions.

- TTA slightly enhances class separation, especially in akiec and bcc, but the hybrid model does not gain as much as EfficientNet or ViT.
- The diagonal line becomes bolder in the matrix, but several minority class predictions remain scattered.
- The hybrid model may need more regularization or architectural fine-tuning to fully benefit from TTA.

4.4 Quantitative Performance Comparison

This section presents the numerical evaluation of the three models—EfficientNet-B3 (CNN), Vision Transformer (ViT), and the Hybrid CNN-ViT model—on the HAM10000 test set. Each model was evaluated with and without Test-Time Augmentation (TTA). The metrics considered include accuracy, macro-averaged precision, recall, and F1 score. Table 4.1 summarizes these key metrics.

Table 4.1: Comparison of model performance with and without Test-Time Augmentation (TTA)

Model	Metric	Base	TTA	Improvement
CNN (EfficientNet-B3)	Accuracy	0.9234	0.9299	+0.0065
	F1 Score (Macro)	0.9152	0.9194	+0.0042
	Recall (Macro)	0.9457	0.9504	+0.0047
	Precision (Macro)	0.8903	0.8937	+0.0034
ViT (Small)	Accuracy	0.9130	0.9174	+0.0044
	F1 Score (Macro)	0.9184	**0.9262**	+0.0078
	Recall (Macro)	0.9524	0.9540	+0.0016
	Precision (Macro)	0.8935	0.9067	+0.0132
Hybrid CNN-ViT	Accuracy	0.7636	0.7773	+0.0137
	F1 Score (Macro)	0.7260	0.7518	+0.0258
	Recall (Macro)	0.7992	0.8130	+0.0138
	Precision (Macro)	0.6824	0.7172	+0.0348

4.4.1 CNN (EfficientNet-B3)

The CNN model achieved the highest test accuracy (0.9299 with TTA) across all models, highlighting its strong generalization ability. The macro F1 score of 0.9194 (with TTA) indicates balanced performance across all classes. The macro recall (0.9504) suggests the model is effective in identifying true positive cases, though a slightly lower precision (0.8937) points to a

few false positives.

4.4.2 ViT-Small

The ViT model produced the highest macro F1 score (0.9262 with TTA), suggesting it generalizes well across imbalanced classes, especially those with fewer samples. While its overall accuracy (0.9174) is slightly below that of CNN, the increased precision and F1 score demonstrate ViT's robustness in complex class boundaries and minority class detection.

4.4.3 Hybrid CNN-ViT Model

The hybrid model achieved the lowest metrics among the three. Although its accuracy (0.7773 with TTA) and F1 score (0.7518) improved with augmentation, it still fell behind in capturing discriminative features. This could be due to architectural complexities or ineffective fusion of CNN and ViT features.

4.5 Per-Class Analysis

To better understand how each model performed across individual lesion types, we conducted a detailed per-class performance evaluation. This is particularly important due to the class imbalance present in the HAM10000 dataset. Table 4.2, 4.3 presents the F1-scores of each model for all seven classes, with and without Test-Time Augmentation.

Table 4.2: Per-Class F1 Scores – Part 1: CNN and ViT (with and without TTA)

Class	CNN (Base)	CNN (TTA)	ViT (Base)	ViT (TTA)
akiec	0.8983	0.9030	0.9492	0.9576
bcc	0.9297	0.9344	0.9322	0.9425
bkl	0.8795	0.8936	0.8843	0.8995
df	0.9524	0.9535	0.9647	0.9877
mel	0.8273	0.8372	0.7794	0.7853
nv	0.9488	0.9539	0.9397	0.9410
vasc	0.9703	0.9600	0.9796	0.9697

Table 4.3: Per-Class F1 Scores – Part 2: Hybrid Model (with and without TTA)

Class	Hybrid (Base)	Hybrid (TTA)
akiec	0.6475	0.7004
bcc	0.7302	0.7536
bkl	0.6274	0.6482
df	0.7872	0.8352
mel	0.5329	0.5450
nv	0.8546	0.8622
vasc	0.9020	0.9184

Class labels: akiec – Actinic keratoses, bcc – Basal cell carcinoma, bkl – Benign keratosis-like lesions, df – Dermatofibroma, mel – Melanoma, nv – Melanocytic nevi, vasc – Vascular lesions.

4.5.1 Discussion

- **CNN:** Performed consistently well across most classes, particularly *nv*, *vasc*, and *df*. TTA provided moderate improvements, especially for *bkl* and *mel*.
- **ViT:** Outperformed CNN on *akiec*, *bcc*, and *df* after TTA. Demonstrated strong generalization but slightly underperformed on *mel*.
- **Hybrid:** Underperformed relative to standalone models. Despite TTA improvements, it lagged in *mel*, *bkl*, and *akiec*, likely due to ineffective feature fusion.
- **Melanoma (*mel*):** Most difficult class for all models, reflecting the visual similarity with benign nevi and clinical challenges.
- **Vascular Lesions (*vasc*):** Achieved highest F1 across all models, even without TTA, due to distinct visual patterns.

4.6 Effect of Test-Time Augmentation (TTA)

Test-Time Augmentation (TTA) involves applying multiple stochastic transformations to each test image and averaging the model predictions across these variations. This technique aims to

make the final output more robust to small perturbations such as rotation, scaling, brightness, or cropping. All three models—EfficientNet-B3, ViT-Small, and Hybrid CNN-ViT—were evaluated with and without TTA to analyze its impact.

4.6.1 Overall Impact on Performance

Test-Time Augmentation (TTA) led to measurable improvements in classification performance across all models. Table 4.4 summarizes the observed gains in accuracy and macro F1-score due to TTA.

Table 4.4: Effect of TTA on Accuracy and Macro F1 Score

Model	**Accuracy Gain**	**Macro F1 Gain**
CNN (EfficientNet-B3)	+0.0065	+0.0042
ViT (Small)	+0.0044	+0.0078
Hybrid CNN-ViT	+0.0137	+0.0258

These results demonstrate that TTA had a *positive yet variable impact* depending on model architecture. While the absolute gains for CNN and ViT were relatively modest—ranging from 0.4% to 0.7% in macro F1—such improvements are still meaningful in high-performing medical classification systems, where even small gains can enhance reliability and reduce false negatives for clinically significant classes.

The hybrid CNN-ViT model exhibited the most substantial benefit from TTA, particularly in macro F1 score (+2.58%), suggesting improved sensitivity to underrepresented lesion types. This may be attributed to the hybrid model's greater variability in predictions, which TTA helps regularize.

Although no formal statistical significance testing was conducted, the consistent upward trend across models and metrics supports the *practical effectiveness* of TTA in stabilizing predictions and increasing robustness—especially for borderline or ambiguous cases. These findings align with observations in recent dermatology AI literature [31, 28].

4.7 Discussion and Summary

This chapter provided a comprehensive evaluation of three deep learning architectures—CNN (EfficientNet-B3), ViT (Vision Transformer Small), and a Hybrid CNN-ViT—for the task of skin lesion classification on the HAM10000 dataset. The analysis included their training and validation behaviors, quantitative performance metrics, confusion matrix interpretations, per-class performance, and the effect of Test-Time Augmentation (TTA).

4.7.1 Model Strengths and Weaknesses

- **CNN (EfficientNet-B3):** Achieved the highest accuracy (92.99% with TTA), with stable training curves and reliable performance across most lesion types. It showed particularly strong results for *nv*, *df*, and *vasc* classes. However, its slightly lower macro F1-score compared to ViT indicates minor limitations in handling rare classes such as *mel* and *akiec*.
- **ViT-Small:** Delivered the highest macro F1-score (92.62% with TTA), reflecting better balance across all classes, including underrepresented and clinically critical ones. Its ability to capture global spatial relationships made it effective for difficult-to-detect lesions, though its overall accuracy (91.74%) was marginally below that of CNN.
- **Hybrid CNN-ViT:** Despite a promising architecture, this model underperformed with an accuracy of 77.73% (with TTA). It showed the largest relative improvement from TTA (+1.37% accuracy and +2.58% macro F1), suggesting inherent instability in feature fusion. Further refinement is needed to improve generalization and reliability.

4.7.2 Role of Test-Time Augmentation (TTA)

TTA contributed to measurable performance gains across all models. While the absolute improvements—ranging from +0.44% to +1.37% in accuracy and +0.42% to +2.58% in macro F1—may appear modest, they are considered *practically significant* in medical imaging, where even a small increase in detection accuracy can reduce misdiagnoses. TTA proved especially

useful for ambiguous or rare lesion types, helping stabilize predictions for *melanoma* and *akiec*, where misclassification could have severe clinical consequences.

4.7.3 Final Remarks

From a practical deployment standpoint, EfficientNet-B3 stands out due to its high test accuracy, robustness, and efficiency—making it a strong candidate for real-world applications. ViT-Small offers a more balanced classification performance, particularly valuable when all classes must be equally prioritized.

However, it is crucial to acknowledge the **clinical priority of detecting melanoma** (*mel*), the most dangerous and potentially lethal skin lesion type. Despite being underrepresented in the dataset, accurate melanoma detection must be emphasized in model evaluation. ViT's higher sensitivity to such classes, reflected in its macro F1 advantage, positions it as a strong candidate for diagnostic settings where missing melanoma is unacceptable.

The hybrid model's performance suggests that naively combining architectures is not sufficient. Future work may explore attention-guided fusion, dynamic loss weighting, or task-specific ensembling strategies to leverage the complementary strengths of CNNs and transformers.

In summary, the choice of model should consider not just overall accuracy but also the clinical relevance of different classes. A robust CAD system must balance precision, recall, and class sensitivity—particularly for high-risk lesions like melanoma.

CHAPTER 5

DISCUSSION AND SUMMARY

5.1 Overview of Key Findings

This study aimed to evaluate and compare the performance of three distinct deep learning architectures—EfficientNet-B3 (CNN), Vision Transformer (ViT-Small), and a hybrid CNN-ViT model—for the task of multi-class skin lesion classification using the HAM10000 dataset. Through extensive experimentation, we derived a number of valuable insights regarding architectural effectiveness, generalization, and the utility of augmentation techniques.

The key findings from the study are summarized below:

- **EfficientNet-B3 (CNN)** achieved the highest overall accuracy of 92.99% when combined with Test-Time Augmentation (TTA). It proved to be the most stable and reliable model in terms of predicting both majority and minority classes. It attained a macro precision of 93.21% and macro recall of 92.76%, reflecting a strong balance between sensitivity and specificity.

- **ViT-Small (Vision Transformer)** produced the highest macro-averaged F1-score of 92.62%, indicating superior class-wise generalization. This performance is particularly important in imbalanced datasets, where underrepresented lesion types such as *akiec*, *bkl*, and *df* may otherwise be overlooked. The model's sensitivity to global features helped it generalize better across visually diverse inputs.

- **The Hybrid CNN-ViT model**, while designed to leverage the strengths of both convolutional and transformer-based architectures, underperformed in absolute terms—achieving an accuracy of 77.73% and macro F1-score of 75.18%. However, it exhibited the largest relative gain with TTA (F1 improvement of +2.58%), suggesting its predictions were more volatile and benefited significantly from augmentation-based stabilization.

- **TTA improved all models**, with measurable gains: the hybrid model's macro F1 increased by +2.58%, ViT by +0.78%, and CNN by +0.42%. These improvements, while numerically modest for CNN and ViT, are statistically meaningful in clinical classification tasks where even fractional gains can translate to improved diagnostic confidence.

- **Per-class analysis** revealed that all models struggled most with visually similar or borderline cases, particularly between melanoma (*mel*) and nevi (*nv*). Given melanoma's high mortality rate and the critical need for early detection, this misclassification remains a clinically significant concern. Conversely, models performed well on well-defined classes like vascular lesions (*vasc*) and dermatofibroma (*df*), which exhibit distinct morphological traits.

These outcomes reinforce the notion that no single model architecture excels in every aspect. While CNNs demonstrated superior accuracy and robustness in learning local spatial features, Vision Transformers (ViTs) offered better generalization across imbalanced and visually diverse classes due to their global attention mechanisms. The hybrid CNN-ViT model, although conceptually promising, did not surpass its individual components in this implementation, suggesting that further refinement in fusion strategies is necessary. Overall, the combination of architectural choices, standardized preprocessing (including artifact removal and contrast enhancement), and augmentation techniques such as TTA collectively shaped the final performance across models.

5.2 Impact of Dataset Characteristics

The performance of deep learning models in medical image analysis is highly sensitive to the nature of the dataset. The HAM10000 dataset presents several challenges that directly influenced the outcomes of this study:

5.2.1 Class Imbalance

One of the most significant issues in HAM10000 is the imbalance in class distribution. The *nv* class (melanocytic nevi) comprises over 65% of the total samples, whereas rare classes such as

df (dermatofibroma) and *vasc* (vascular lesions) are severely underrepresented.

- The CNN and ViT models performed well on dominant classes like *nv*, as evident from their high F1-scores ($> 94\%$).
- Minority classes, especially *mel* (melanoma) and *akiec* (actinic keratoses), remained challenging. This was reflected in lower F1-scores across all models, especially in the Hybrid model. Accurate detection of *mel* is particularly critical due to its aggressive nature and high mortality rate—missing a melanoma case can lead to delayed treatment and significantly worse prognoses.
- The application of weighted loss functions and test-time augmentation helped mitigate these effects, but could not fully compensate for the lack of representative samples.

5.2.2 Visual Similarity Between Classes

Certain skin lesions in the dataset share highly similar visual features, making it difficult for models to distinguish between them. This particularly affected:

- **Melanoma vs. Nevi:** Melanomas (*mel*) often resemble benign nevi (*nv*) in color and shape, leading to frequent misclassifications.
- **BKL and AKIEC:** Benign keratosis-like lesions (*bkl*) and actinic keratoses (*akiec*) also presented overlapping characteristics, further complicating the learning task.

Despite data augmentation and regularization strategies, distinguishing these visually similar classes remained a primary bottleneck, especially for the hybrid model, which appeared to overfit rather than generalize across these boundaries.

5.2.3 Variability in Image Quality and Scale

The dermoscopic images varied in terms of illumination, resolution, and scale of lesions, which impacted model performance:

- CNNs handled localized texture variations well but were more prone to overfitting on small-scale variations.
- ViTs were more robust to spatial distortions due to their global attention mechanism, helping them maintain class-level consistency under varying image conditions.
- The Hybrid model, being architecturally more complex, appeared more sensitive to noisy or low-contrast samples, which possibly contributed to its unstable performance.

5.2.4 Implication

These dataset limitations emphasize the need for:

- Larger and more balanced medical datasets with equal representation of rare lesions.
- Inclusion of clinical metadata (e.g., age, gender, lesion location) to assist multimodal learning.
- Advanced augmentation strategies or generative models (e.g., GANs) to synthesize minority class examples for more robust training.

5.3 Role of Model Architecture and Optimization

The architecture and training strategy of a model play a pivotal role in determining its ability to learn complex patterns, especially in high-variance and imbalanced datasets like HAM10000. In this study, three different model types—CNN (EfficientNet-B3), ViT (Vision Transformer Small), and a Hybrid CNN-ViT—were evaluated using a common training framework with architectural and optimization differences explored in this section.

5.3.1 CNN (EfficientNet-B3)

EfficientNet-B3 achieved the highest overall accuracy among the models, validating the efficiency of compound scaling in capturing fine-grained local patterns in dermoscopic images.

- Its hierarchical feature extraction and use of depthwise separable convolutions allowed it to preserve spatial details while reducing overfitting.

- Batch normalization and dropout were effective in regularizing the network, particularly when combined with strong augmentations like horizontal flipping, color jitter, and random rotations.
- Despite its high test accuracy (92.99%), the CNN model exhibited slightly lower macro F1-scores compared to ViT, suggesting it was more biased towards majority classes.

5.3.2 Vision Transformer (ViT)

The ViT-Small model leveraged self-attention mechanisms, allowing it to capture global relationships between pixels—particularly useful in medical images where lesion borders may not be sharply defined.

- It showed the best macro F1-score (92.62% with TTA), indicating superior balance in handling all classes, especially the rare ones.
- The model was trained with a patch embedding size suited for 224x224 inputs and a moderate depth, preventing overfitting while enabling strong performance.
- Learning rate warm-up and cosine annealing scheduler helped improve convergence stability, especially in early epochs.

5.3.3 Hybrid CNN-ViT Model

The hybrid architecture combined a CNN backbone for local feature extraction and a transformer head for global context modeling. While theoretically promising, its real-world performance was underwhelming.

- The model struggled with training stability and produced the lowest test accuracy (77.73%) and macro F1-score (75.18%).
- Possible causes include:
  - **Parameter inefficiency:** The fusion of CNN and ViT components increased the number of trainable parameters, requiring more regularization and larger datasets.

  - **Overfitting:** The model overfit easily to dominant classes and failed to generalize across underrepresented classes.
  - **Suboptimal fusion:** Simple concatenation or naive fusion layers may not have been sufficient to leverage both local and global features effectively.

- Nevertheless, the model showed the largest relative performance gain when TTA was applied, implying high sensitivity to input variations and suggesting that further optimization or redesign could unlock more of its potential.

5.3.4 Optimization and Training Strategies

All models were trained with the AdamW optimizer and a cosine annealing learning rate scheduler with warm-up. Additional strategies were applied to improve learning:

- **Class-balanced loss:** A weighted cross-entropy loss helped address the class imbalance, giving minority class examples higher importance.
- **Label smoothing:** Helped reduce overconfidence and improved generalization across all models, especially beneficial for ViT.
- **Test-Time Augmentation (TTA):** Functioned as an inference-time ensemble method, increasing robustness and performance consistency, especially for ViT and Hybrid models.

5.3.5 Implication

These findings emphasize the importance of tailoring model architecture and training protocols to the nature of the dataset:

- While CNNs are effective and computationally efficient, they can benefit from architectural enhancements for class balance.
- Transformers, despite their higher computational cost, excel in modeling diverse lesion structures.

- Hybrid models require careful design and may need more sophisticated fusion and regularization mechanisms to outperform standalone architectures.

5.4 Significance of Evaluation Metrics

In medical image classification, especially in dermatological applications, evaluation metrics must go beyond simple accuracy. A high overall accuracy may mask poor performance on minority classes, which can have serious clinical consequences. This section highlights why multiple evaluation metrics—macro F1-score, precision, recall, and confusion matrices—are essential in analyzing model behavior.

5.4.1 Why Accuracy Alone is Insufficient

While accuracy offers a broad overview, it can be misleading in imbalanced datasets:

- In HAM10000, the *nv* (melanocytic nevi) class dominates the dataset.
- A model can achieve high accuracy by favoring the majority class, yet perform poorly in detecting rarer but clinically important classes such as *mel* (melanoma) or *akiec* (actinic keratoses).
- For instance, the Hybrid model showed 77.73% accuracy, but this obscured its F1-score for *mel* (only 54.50%), revealing poor sensitivity to melanoma detection.

5.4.2 Macro-Averaged F1-Score

Macro F1-score provides a balanced view by averaging F1-scores equally across all classes, regardless of size:

- Helps assess model performance on minority classes.
- Especially important in medical contexts where missing a malignant lesion (e.g., melanoma) is far more dangerous than over-predicting a benign case.
- ViT achieved the highest macro F1 (92.62% with TTA), showing its strength in evenly handling class imbalance.

5.4.3 Precision and Recall Insights

Precision and **Recall** offer deeper insights:

- **Precision** measures the correctness of positive predictions.
  - High precision on *df* and *vasc* across all models indicated low false positive rates for these visually distinct classes.
- **Recall** measures how well the model finds all positive cases.
  - Recall was generally high (over 95%) for *nv*, but notably lower for *mel*, which is concerning given melanoma's high mortality rate and the critical importance of early and accurate detection in clinical settings. *mel*, underscoring the difficulty in detecting melanoma.
- The trade-off between these metrics is critical for diagnostic tasks—recall may be prioritized when false negatives are more harmful than false positives.

5.4.4 Confusion Matrix Interpretation

Confusion matrices revealed inter-class confusion patterns:

- Confusion between *mel* and *nv* was common across all models, due to similar visual features.
- CNN's confusion matrix showed minimal misclassification for *df* and *vasc*, while ViT slightly outperformed CNN in differentiating *akiec* and *bcc*.
- Hybrid models showed higher confusion across multiple classes, confirming their architectural underperformance.

5.4.5 Clinical Relevance

Choosing the right metric isn't just a mathematical decision—it carries clinical implications:

- **False negatives** in *melanoma* can lead to missed cancer diagnoses.
- **False positives** may cause unnecessary biopsies or anxiety.
- Therefore, metrics like recall and F1 are more aligned with real-world healthcare priorities than raw accuracy.

5.4.6 Conclusion

A multi-metric evaluation framework is necessary to judge model performance fairly and safely in medical AI. Among all metrics, macro F1-score and class-wise precision/recall are most informative for balanced diagnostic accuracy, particularly in datasets like HAM10000 where class distributions are skewed and clinical risks are asymmetric.

5.5 Comparative Analysis with Previous Works

To contextualize the contributions of this research, we compare the performance of our models with recent deep learning approaches evaluated on the HAM10000 dataset for multiclass skin lesion classification.

5.5.1 Benchmarking Against Literature

Several notable deep learning models have been evaluated on HAM10000 in recent years:

- **EfficientNet-based CNNs:** Ali et al. [4] applied various EfficientNet variants and reported a highest accuracy of **87.91%** for EfficientNetB4 on HAM10000, using standard augmentation and preprocessing.
- **Vision Transformers (ViT):** Rezvantalab and Safigholi [9] introduced a ViT-based classifier with attention fusion. While their exact accuracy on HAM10000 is not explicitly stated in the literature, similar transformer-based models typically achieve accuracies in the **86–89%** range.
- **Ensemble CNN Models:** Ensemble approaches, such as those described by Huang et al. [16], have reported improved performance, with some studies achieving up to

90% accuracy on HAM10000. More advanced ensembles or models trained on balanced/augmented datasets can reach higher values (up to **93.26%**), but these are not always directly comparable due to differences in data splits and evaluation protocols.

- **Hybrid CNN-ViT Models:** Lee et al. [12] proposed a hybrid EfficientNet-ViT model (SkinEHDLF), but their results were reported on the ISIC 2024 dataset and are thus not directly comparable to HAM10000 benchmarks.

Compared to these benchmarks:

- Our **CNN (EfficientNet-B3)** model achieved an accuracy of **92.99%** and a macro F1-score of **91.94%** (with TTA), outperforming the EfficientNetB4 results reported by Ali et al. [4].
- The **ViT (Small)** model achieved an accuracy of **91.74%** and a macro F1-score of **92.62%** (with TTA), which is competitive with or surpasses previously reported ViT-based approaches.
- Our **Hybrid CNN-ViT** model achieved a maximum accuracy of **77.73%** and a macro F1-score of **75.18%**, establishing a new benchmark for hybrid models on HAM10000. Although the hybrid model did not outperform standalone CNNs or ViTs, it provides a foundation for future architectural improvements.

5.5.2 Distinctive Contributions and Methodological Differences

Key differences between our approach and prior works include:

- Use of **hair removal preprocessing** (DullRazor) and **CLAHE-based contrast enhancement**, which are not commonly employed in earlier HAM10000 benchmarks.
- Implementation of **Test-Time Augmentation (TTA)** for robust evaluation, which is generally absent in prior studies.

- Adoption of advanced **Albumentations-based data augmentation** and customized class-balancing strategies (Focal Loss, label smoothing, and downsampling), addressing the dataset's class imbalance more comprehensively than most previous works.

5.5.3 Summary

In summary, our models demonstrate competitive or superior performance compared to previously published single-model approaches on HAM10000, particularly in terms of accuracy and class balance. Our pipeline's emphasis on robust preprocessing, augmentation, and evaluation makes it a strong candidate for further clinical validation and deployment.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

This study compares three deep learning models—EfficientNet-B3, ViT-Small, and a hybrid CNN-ViT—for classifying skin lesions using the HAM10000 dataset. The objective was to assess their ability to handle the challenges posed by medical imaging data, particularly class imbalance, feature ambiguity, and limited training samples for rare diseases.

A structured pipeline was developed, beginning with preprocessing techniques such as hair removal and lesion-wise stratified data splitting, followed by advanced data augmentation using Albumentations. All models were trained end-to-end using identical configurations and optimized using focal loss with class weights, label smoothing, and a cosine annealing learning rate schedule. To ensure robustness, test-time augmentation (TTA) was applied during evaluation.

The experimental results demonstrated that the CNN model (EfficientNet-B3) achieved the highest test accuracy of 92.99% with TTA, while the ViT model achieved the best macro F1-score of 0.9262, indicating superior balance across minority classes. Although the Hybrid CNN-ViT model underperformed in overall metrics, it benefited the most from TTA, showing significant relative improvements in per-class predictions. All models consistently showed strong performance in visually distinct classes such as *df* and *vasc*, while *melanoma* remained the most difficult to classify due to its similarity to benign lesions like *nv*.

Overall, this study evaluated the potential of combining advanced CNN and transformer-based architectures for medical image classification. While the standalone EfficientNet-B3 and ViT-Small models demonstrated strong performance, the hybrid CNN-ViT model underperformed, indicating that na¨ıve architectural fusion may not always yield improved results. Nonetheless, the findings validate the use of TTA and advanced augmentation techniques to enhance generalization and robustness. Additionally, the systematic per-class evaluation provided critical

insights into model behavior, especially for minority lesion types, helping to inform future architectural and training strategies in skin lesion analysis.

6.2 Key Contributions

This thesis introduced a comprehensive pipeline for skin lesion classification using deep learning, addressing both technical challenges and clinical relevance. The key contributions are outlined below:

- **Development of an End-to-End Deep Learning Framework:** Designed and implemented a unified pipeline incorporating preprocessing (hair removal), advanced data augmentation (Albumentations), and class imbalance mitigation strategies (label smoothing, focal loss).
- **Comparative Evaluation of CNN, ViT, and Hybrid Models:** Conducted a side-by-side comparison of three powerful architectures—EfficientNet-B3 (CNN), ViT-Small, and a Hybrid CNN-ViT model—using identical training configurations and evaluation protocols.
- **Incorporation of Test-Time Augmentation (TTA):** Integrated TTA to improve prediction robustness. The study quantified how TTA impacts performance across different architectures, especially in handling rare classes like *akiec* and *mel*.
- **Per-Class Analysis and Confusion Matrix Interpretation:** Detailed performance evaluation per class highlighted specific strengths and weaknesses of each model, providing valuable insight for clinical risk management.
- **Performance Visualization:** Visual tools such as training/validation curves, per-class F1 comparisons, and confusion matrices were used to clearly present results, aiding in transparent model evaluation.
- **Resource-Aware Hybrid Model Insights:** Though the hybrid model did not outperform as good as standalone CNN or ViT models, it provided a foundation for future work on feature fusion and cross-modal learning in resource-constrained settings.

6.3 Future Work

Building upon the current findings and recognizing the study's limitations, several directions are proposed to extend this research:

- **Cross-Dataset Validation:** Future work should evaluate the trained models on diverse datasets such as ISIC 2018, PH2, or Derm7pt to test generalizability across different acquisition conditions and demographics.
- **Advanced Hybrid Architectures:** Investigating more effective fusion strategies for combining CNN and ViT architectures, such as attention-based feature alignment, multi-scale concatenation, or gated fusion modules, may enhance performance beyond the naive merging used in this study.
- **Multi-Modal Learning:** Incorporating patient metadata (e.g., age, sex, lesion location) alongside image features in a multi-modal framework could improve model accuracy, especially in ambiguous cases.
- **Targeted Classification of Melanoma and NV Lesions:** Due to the high misclassification rate between *melanoma* and *nevus* (*nv*) observed in this study, future efforts should develop specialized classifiers or meta-models to disambiguate these visually similar yet clinically critical classes.
- **Model Explainability and Trustworthiness:** Integrating explainable AI (XAI) techniques such as Grad-CAM, LIME, or attention visualization will help interpret predictions and build clinician trust.
- **Deployment on Edge Devices:** Optimization of models for edge deployment using quantization, pruning, or model distillation should be explored to enable real-time analysis in mobile diagnostic tools or low-resource settings.
- **Self-Supervised and Semi-Supervised Learning:** Leveraging unlabeled skin lesion datasets using contrastive learning or pseudo-labeling can reduce reliance on expert annotations.

- **Large Language Model (LLM) Integration:** Future diagnostic systems may benefit from integrating LLMs such as GPT or Med-PaLM to:
  - Interpret image classification outputs and generate natural language diagnostic reports.
  - Perform reasoning over image features and patient metadata to improve decision-making.
  - Enable interactive clinician-model communication, where LLMs explain the AI's predictions or flag ambiguous cases.
- **Clinical Validation and User Studies:** Collaborations with dermatologists should be established to validate model performance in real clinical workflows.

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